Module 3

Non-digital and Digital Skills and Tools in Delivering Technology-Enhanced Lesson

Teaching becomes rewarding when learners get the most from instruction as manifested in their performance. An important element in engaging learners is when the strategy used in delivering the lesson uses an instructional material. When properly and appropriately used, it can spice up a classroom activity.

These instructional materials may come in varied forms. One group refers to the conventional and non-digital tools. A classroom will always need a chalkboard or a writing board that may come in varied forms and shapes. Bulletin boards, flip charts, dioramas, puppets, terrarium, and like, will always find their significance in any classroom. However, nowadays, lessons can be made more relevant and engaging for learners as digital tools are integrated. This module presents both non-digital and digital tools. Explore the possibilities of learning about these tools and how to effectively integrate them in instruction.

Lesson 1

Development and Use of Non-Digital or Conventional Materials

Lesson Outcomes

1. Describe the procedures for developing conventional instructional materials
2. Described the factors to consider in revising media selections and delivery systems for given instruction

Introduction

The teachers need instructional materials to enhance teaching and learning. Instructional materials are defined as print and non-print items that are rested to impact information to students in the educational process (Effiong & Igri, 2015). Examples of instructional materials are drawings, kits, textbooks, posters, magazines, flip chart, newspapers, diorama, pictures, recording videos and the like.

Instructional materials have several roles in teaching and learning which include the following: (1) they promote meaningful communication and effective learning; (2) they ensure better retention, thus making learning more permanent; (3) they help to overcome the limited classroom by making the

inaccessible accessible; (4) they provide a common experience upon which late learning can be developed; and (5) they encourage participation especially if students are allowed to manipulate materials used (Brown et al., 2005; Effiiong & Igiri, 2015).

Discussion

Instructional materials are the supplementary materials, which helps the teacher to make his/her presentation concrete, effective, interesting, meaningful and inspiring. In any teaching and learning process, instructional materials play a vital role as they provide sensory experiences to the learners. The primary aim of teaching materials is to provide the teachers the layout of the way for teaching in the classroom.

It is important to understand how to develop instructional materials. Instructional materials refer to any preexisting materials that are being incorporated, as well as to those that will be specifically developed for the objectives (Haigler, 2014). There are also several factors to consider in developing instructional materials:

1. Develop a story board and working outline based on the subject goals and objectives. 2. Identify existing instructional resources including materials and teachers' capability 3. The teacher may research off the shelf materials that have been developed by the others to determine if their approach could be useful.
4. Explore the possibility of adapting concepts of other teachers without infringing on anyone's copy protected design.
5. Modify existing materials based on the objectives of the lesson.
6. If the instructional materials are effective, you can share them with other teachers.
7. The teacher developer can also sell his/her materials available.

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Technology for Teaching and Learning 1

1. Diorama



Dioramas are small scenes created of layers of materials, all depicting a similar concept or theme. They usually display a historical time period, a nature scene, or a fictional situation. In developing diorama, you will: (1) choose a concept or theme, (2) research the subject, (3) make a rough sketch of your ideal diorama, (4) make a list of the items you'll need and gather your supplies, and (5) select a container or a box.

2. Nature Table



This is a table that contains objects and/or scenes related to the current season, or upcoming festival or a symbol of an ecosystem. Children love to follow the natural

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3. Writing Board



A writing board can display information written with chalk (chalkboard or blackboard) or special pens (whiteboard).

Suggestions on Using the Writing Board:

1. Keep the board clean.
2. Use chalk or pens that contrast with the background of the board so that students can see the information clearly.
3. Make text and drawing large enough to be seen from the back of the room.
4. Prepare complex drawings in advance (if every complex, an overhead transparency or 35 mm slide may be preferable).
5. Underline headings and important or unfamiliar words for emphasis.
6. Do not talk while facing the board.
7. Do not block the students' views of the board; stand aside when writing or drawing is

completed.

8. Allow sufficient time for students to copy the information from the board.

4. Flip chart



It is a large tablet or pad of paper, usually on a tripod or strand.

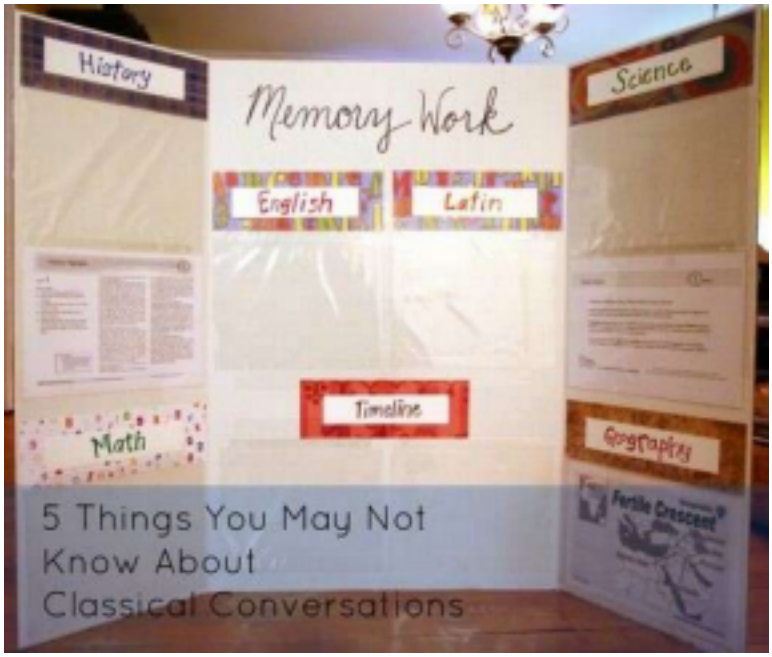
Suggestions on Using Flipchart:

1. Use wide-tipped pens or markers; markers with narrow tips produce printing that is difficult to read.
2. Print in block letters that are large enough to be read easily from the back of the room.
3. Use different colored pens to provide contrast; this makes the pages visually attractive and easier to read.
4. Use headings, boxes, cartoons and borders to improve the appearance of the page.
5. Use bullets to delineate items on the page.
6. Leave plenty of "white-space" and avoid putting too much information on one page. (Crowded

and poorly arranged information is distracting and difficult to read).

7. When pages are prepared in advance, use every other page. If every page is used, colors will show through and make text difficult to read.
8. Have masking tape available to put pages up around the room during brainstorming and problem – solving activities.
9. To hide a portion of the page, fold up the lower portion of the page and tape it. When ready to reveal the information, remove the tape and let the page drop.
10. Face the student, not the flipchart while talking.

5. Zigzag board



It is a multi-board series of three or four rectangular boards. They are joined together along the sides by hinges so they can be easily folded up and carried. Each board can be of a different type, for example, a whiteboard, a chalkboard, a flannel board and so on.

6. Wall display



Displaying items on a classroom wall is a well-known, tried and tested educational method. A wall display is a collection of many different types of items and materials put up on a wall to make an interesting and informative display. In a classroom, the display can consist of the students' own work. In development work it can be used to convey information to the community.

7. Rope and Pole display board



This board consists of two parallel, horizontal poles tied loosely together with rope. Visual aids such as posters can be pinned to the rope. This kind of display board is invaluable where there are few solid walls for displaying information. It has no solid backing and can be made quickly for teaching, training and when working with communities.

Guidelines when designing conventional instructional materials:

1. Unity – Use only one idea for each visual aid and include a headline.
2. Simplicity – Makes ideas and relationships simple and easy to recall. Avoid cluttering a visual with too many words, numbers, or graphics. The audience should be able to grasp the concept in 10 to 15 seconds.
3. Legibility – Make letters big and readable for all in the audience.
4. Consistency – Use the same type style and art style.
5. Clarity – Avoid type that is too small to read; avoid all caps
6. Quality – Make it neat and professional, and remember to proofread.

To Do!

Learning Activity 3.1. Discuss the importance of the following in the teaching and learning process.

1. Flipchart
2. Wall Display
3. Chalkboard
4. Diorama
5. Nature Table

Reflective Essay 3.1.: In writing your reflection always uses Gibb's Reflective Model. Reflect also in

this concept. Instructional materials compromise a wide variety of items, handouts, slides moving pictures, moving pictures, posters, models, objects and many others. All of these are meant to enhance the teaching and learning process. Do you think, non-digital instructional materials are still relevant in the 21st century teaching and learning. Agree or Disagree?

Lesson 2

Creating ePortfolio as a Technology Tool

Lesson Outcomes

1. **To familiarize in creating ePortfolio**

Introduction

In the 21st century instruction where independent learning is encouraged; the documentation of a personal learning journey is a must. Such documentation can help the teacher monitor the process and areas the product of learning. Doing this requires proper organization through a portfolio.

Portfolios are one good means of keeping things in order. In a semester work, a learning portfolio is a collection of students work that exhibit students' effort, progress, achievements and competencies gained during the course.

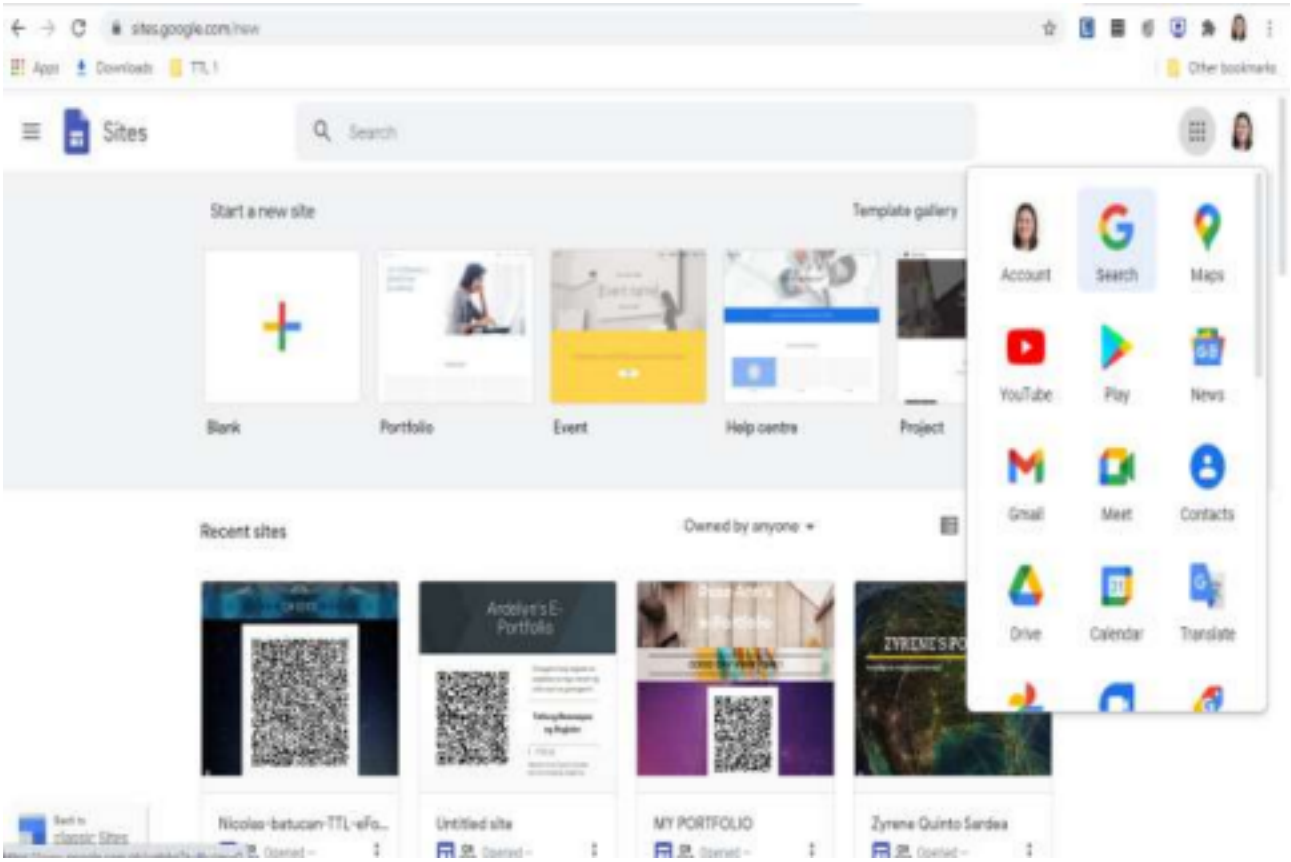
Portfolios may come in many forms. It can look like an album or scrapbook or even a filer where the documents and evidences are kept. However, nowadays, it is possible to have online portfolios by creating sites. This is also called the ePortfolio or digital portfolio. This can be used as a digital archive that can contain the same materials as a physical portfolio but can have more such as multimedia productions, relevant online links or references, digital stories or video blogs, PowerPoint presentations, photographs and other ICT materials. The ePortfolio can be private or can be published and shared publicly to stakeholders like parents and friends.

1. Student ePortfolios can evaluate students' academic progress. They can inform the teacher to adapt and use instructional strategies when pieces of evidences indicate that they are either learning or not.
2. Monitoring students' progress can be highlighted in a portfolio. It may not only contain finished products but also several versions on how the students improved their work based on the feedback provided by mentors.
3. Portfolios document students' learning growth. They actually encourage the students' sense of accountability for their own learning process. This may lead them to see that the learning process is theirs and not anybody else's.

Creating an Online Portfolio Using a Site

There are many sites that can be used in creating an ePortfolio. One of which is the google site. If you have a google account, you can start using the available applications. You can also try weebly, or wix, among others.





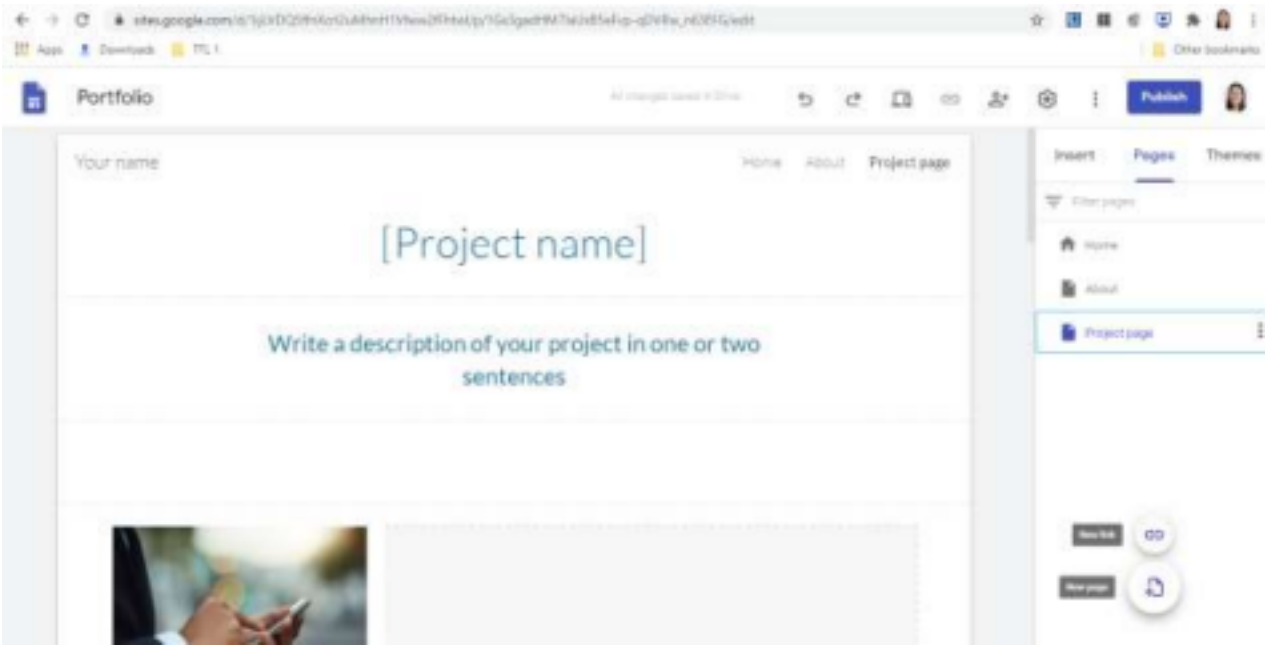
Steps in constructing an ePortfolio

1. Enter your gmail account and look for Sites. If it is the first time that you have done this, you need to read the directions.
2. You scroll down and read further until you see the icon for Sites.
3. When you click it, it will lead you to another section. This will let you create a site that you can use as an ePortfolio.
4. Consider a good label or a title for your ePortfolio and prepare the texts, links multimedia outputs, images or jpeg files that you want to upload in the pages of the ePorfolio.

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Parts of an ePortfolio

Just like a book, the ePortfolio has pages or sections. The organization can follow a chronological order based on the activities that you go through or you can have a thematic arrangement. Whatever you choose, it will be a display of your organizational skills.



Home Page

The first section is the Home or your cover page. This is the first thing that your ePortfolio. Usually, there are templates available and each provides sections. You can add personal touches such as images or a change of color themes.

Pages

The pages that you can add depend on how you would like to organize your ePoretfolio. What is important is that you need to construct your ePortfolio at the start of the class. In that way, that way, you can have a fresh start as you try to be conscious in documenting the activities and learning that goes with each session.

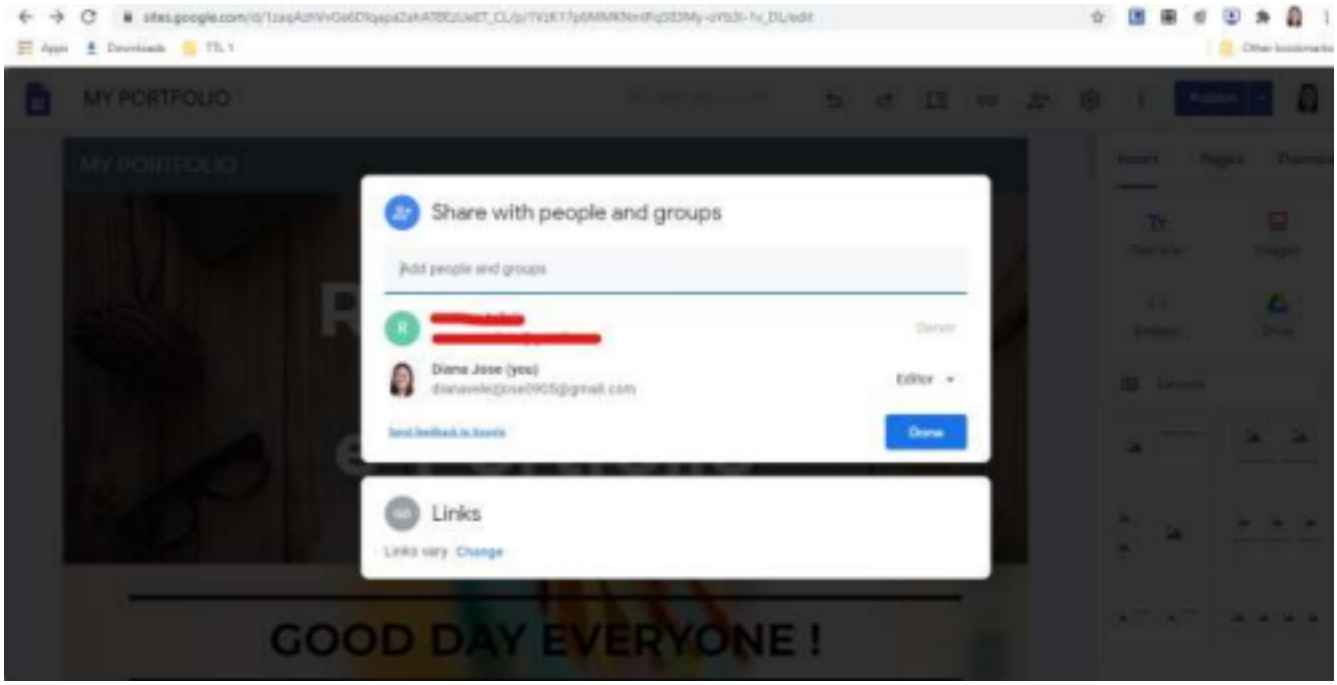
When adding pages, click the icon and decide whether it will be parallel to your Home Page or it will be under it. Just remember that when you have a template, there are particular ways that the

pages have been arranged. So if you are starting, it would not be detrimental if you conform to

the template.

Administering the Portfolio

Befor4e publishing your ePortfolio for the world to see your work, you can control who can see your work. The icon for sharing the site can be managed by entering the email address of the person with whom you want to share it.



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Lesson 3

Digital Literacy Skills in the 21st Century

Lesson Outcomes

1. Identified and described the elements of digital literacies
2. Conducted a self-evaluated on level digital literacy skills

Introduction

The 21st Century has redefined digital literacy. It has broadened its perspective to include other aspects of the 21st context. These literacies include (1) Cyber Literacy or Digital Literacy, (2) Media Literacy, (3) Arts and Creativity Literacy, (4) Financial Literacy, (6) Multicultural Literacy or Global Understanding.

The millennial students are generally tech-savvy, digital natives. They practically know how to go about a tablet, an ipad, smartphones or laptops better than anyone else. This suggests their digital skills.

However, are all students' digital literates?

Thinking on how to use digital tools, accessing information, and exhibiting ways of working in a globally competitive contexts, together with skills in living in the 21st Century refer to digital literacies.

Self-Evaluation

Do you feel confident when you search for information in the internet? Do you easily locate relevant resources? Do you think you are capable of optimizing the use of the online tools such as Facebook or google when studying or doing school work? How do you manage your digital identity?

The skills you need to be able to perform well in situations are known as "**digital literacy**". In the 21st century, you need to be equipped with these skills to be abreast with the demands in the workplace.

Take the **Digital Literacy Skills Self-Inventory** and fill out how confident you are when working online by putting a tick mark [√] in the corresponding column. Note: Do not include this for submissions.

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Digital Literacy Skills- Finding, Using, Creating Information and Understanding Digital Practices	**Not Confident**	**Somewhat Confident**	**Very Confident**
Selecting the right tool when you need to locate, use or present information			
Presenting your digital identity online			
Identifying and communicating with an expert online			
Using online tools and sites to search or share information online			
Recognizing owner of ideas and information found online through citations			
Distinguishing the search engines to get information			

Knowing the information that can be searched in the web			
Using the right key words to search for related online materials			
Using social networking sites as a source information			
Scanning or skimming a page of a site or a blog among others to get the relevant information quickly			
Signing up in pertinent sites having experts and authorities in the discipline to be updated with information they can provide			
Using materials in a variety of online media (i.e. videos, podcast, sites among others)			
Accessing the credibility of an online resource			
Citing correct reference to other person's work found online			
Using bookmarking to organize and legally share files			
Posting comments to online forum, blogs or Vblogs, or web pages while observing netiquette			
Writing and posting online views or ideas suited to the target readership or audiences			
Collaborating with others in creating and sharing document or presentations			
Using multimedia to capture the information or message (i.e. recording and editing a digital story, video or a podcast)			
Communicating with others through online modality (e.g. skype, forums, blogs, social networking sites, etc.)			

Discussion

What are digital literacies?

Digital literacies are the individual's capabilities to be able to effectively and responsibly function and perform in a digital society. The term "digital literacy" was coined by Paul Gilster in 1997 and it

came from the discussion of the concepts on (a) visual literacy when images and non-verbal symbols try to capture knowledge; (b) technological literacy requiring one to be able to use technology in addressing a need (c)computer literacy, which in the 1980s started to become a household item manipulated to achieve one's target; and (d) which refers to the finding, evaluating, using and sharing of information.

In the teaching and learning context, digital literacy is an important competence. In school, it has become a buzzword which refers to the ability to access, process, understand, utilize, create media content using information technologies and the internet (Hsieh,2012). Study the varied digital literacies discussed below.

The Digital Literacies

Media Literacy _ is one's ability to critically read information or content and utilize multimedia in creatively producing communications.

Information Literacy – is locating information from the web and interpreting while evaluating its validity in order that it can be shared.

ICT Literacy – is knowing how to select and use digital devices, applications or services to accomplish tasks requiring the use of the internet.

Communications and Collaboration – are one's capabilities in being able to participate in the digital networks in the teaching and learning context.

Identity Management - is being able to understand how to ensure safety and security in managing online identity and foster a positive digital reputation.

Learning Skills – are ways of knowing how to study and learning a technology-enriched environment; this is knowing how to utilize technology in addressing the need to learn efficiently.

Digital Scholarship – is being able to link and participate in professional and research practices

One important component of digital literacy is having an in-depth understanding of concepts

requiring essential core skills. These core skills are known as the Cs of the 21st Century Skills.

The Four Cs of the 21st Century Skills



The four Cs of the 21st Century Skills refer to **critical thinking, creativity , communication** and **collaboration.** To perform well in this century, you need to develop and enhance these skills namely creativity, critical thinking, communication and collaboration.

1.Critical Thinking is learning how to solve problems. It teaches students not to accept immediately claims without seeking the truth. It is the ability to differentiate facts from opinions and not only just learn a set of facts or figures but also discover these for the sake of knowing what ought to be.

2. Creativity requires students to think out of the box and to take pride in what is uniquely theirs. IT means that they will be able to look at a problem from multiple perspectives- including can propose multiple possibilities and alternatives to address a problem and they need to take calculated risks. Creativity encourages students to think beyond expectations of conventions. However, creativity ma not ensure success all the time but it may lead to another direction that can actually be a better way of figuring out how to solve the problem those that others may not see.

3. Communication make students express their ideas in the clearest and organized manner. Through varied modes – face-to-face, technologically mediated or a blended medium, they need to

know how to efficiently and clearly convey ideas.

4. Collaboration happens when students know how to work well with others to accomplish a given task or solve a problem at hand. This is a 21st century essential. When students are made to work with others in a pair or in a team, they are given the chance to practice how to relate with others. They may be working with classmates they do not really prefer to work with but with guidance, you can teach them to tap on the capabilities of each member of the team and collaboratively achieve the goal. Eventually when they leave school, they will definitely see the need to know how to collaborate with others in order for them to accomplish a job, and you have prepared them for it. In other words, collaboration teaches students that groups can create something bigger and better than you can on your own.

In addition to the 4Cs, there are **Citizenship** and **Character**. Citizenship is known as netizenship in the virtual world. This is making the person consider how one behaves accordingly by observing the norms and rules that are in accordance with what are sociably and virtually acceptable. As a result, one is projecting a reputable digital identity which is his or her character.

Digital Literacy Skills vs. Digital Literacy

Digital Literacy Skills are required in the wired world. These skills vary from text to images to multimedia. Future teachers who will be handling students considered as tech savvy should equip themselves with competencies and fluencies needed to handle the fast emerging tools and applications that should be able to handle even artificial intelligences.

Lynch (2017), identified eight digital literacy skills needed to become digitally literate. These are:

1. **Coding** – Coding is a universal language. Basic understanding of HTML,CSS and like will create a shared understanding of what can be done with web pages.
2. **Collaboration** – The use of Google Docs among others allows student to begin experimenting with effective online collaboration.
3. **Cloud software** – this is essential part of document management. The cloud is used to store everything from photos to research projects, to term papers and even music.

4. **Word Processing Software** – Google, Microsoft Online Drop Box are available for storage

and management solutions

5. **Screencasting** – A screencast is a video recording using the computer screen and usually includes an audio. On the other hand, when you take a picture on the screen of your computer, it is called Printscreen. Both can be used in explaining topics as well as providing a visual support to clarify what you are thinking. While the printscreen is a picture, the screencasting is a video screen capture which is a great way to share ideas and is easy for use for novice video creator.
6. **Personal Archiving** – Students should be taught the concepts of metadata, tagging, keywords and categories to make them aware how are they represented online

7. **Information evaluation** – Critical thinking to weed out fake news is a crucial 21st century skill. The use of tools and skills needed to process information are very much needed. 8. **Use of social media** - Social media serves different purposes depending on the user, the technology and the need. For example, student should realize that Twitter can be useful for staying current on the latest new in the field.

Digital literacy has been defined in many ways that is understandable by both digital natives and digital immigrants. Teachers and students should not only be proficient in how to use (digital skills) but they need to see the information and media technology to find, evaluate, create and communicate information requiring both cognitive and technical skills.

Here are examples of how Digital Skills or Proficiency support Digital Literacy.

Digital Skills	**Digital Literacy**
1. Sending an email or text	1. Evaluate the appropriate digital channel for online communication with peers, teachers and parent
2.Using Microsoft office/ Google	1. Identify the benefits and drawback of each digital tool 2. Evaluate critically which tool is most effective for the project at hand
3. Tweeting, Posting to Facebook, Uploading a Video to YouTube or posting a photo to Instagram	1. Navigate the social media safely to protect oneself 2. Identify hate propaganda and fake news
4. Researching from the worldwide web	1. Evaluate the information online a. Is the site legitimate?

	b. Is the author an expert? c. Is the information current or dated? d. Is the idea neutral or biased?

To Do!

Learning Activity 3.3: Apply your digital literacy skills on information literacy by searching for the identified sample sites. Navigate around each site and share what information can be found in each. Share how you will make use of the information or content that you can retrieve from those sites.

1. Professor Garfield http://www.professorgarfield.org/parents_teachers/home.html 2. Common Sense Media https://www.commonsense.org/education/digital-citizenship 3. Lesson Plan Boosters: Digital Literacy and Online Ethics https://www.educationworld.com/a_lesson/lesson-plan-booster/cyber-ethics.shtml

Reflective Essay 3.3: In writing your reflection always use Gibb's Reflective Model and discuss the concept below:

Based on the result of your Digital Literacy Skills Self-Inventory, in which following area (s) do you find yourself least confident?

a. Understanding digital practices
b. Locating information
c. Using information
d. Creating Information

Module 4

Theories and Principles in the Use and Design of technology-Driven

Lessons Introduction

How teachers integrate technology in the teaching and learning process depends very much on their beliefs on how people learn. Specifically, they need to know who their learners are and how to approach instruction. As educators, their role is to provide learning experiences that will help achieve the defined outcomes. In this Module you will be acquainted with different theories and learning principles such as Dale's Cone of Experience, the TPACK Framework, SAMR and the ASSURE Model.

Lesson 1

Edgar Dale's Cone of Experience

Lesson Outcomes

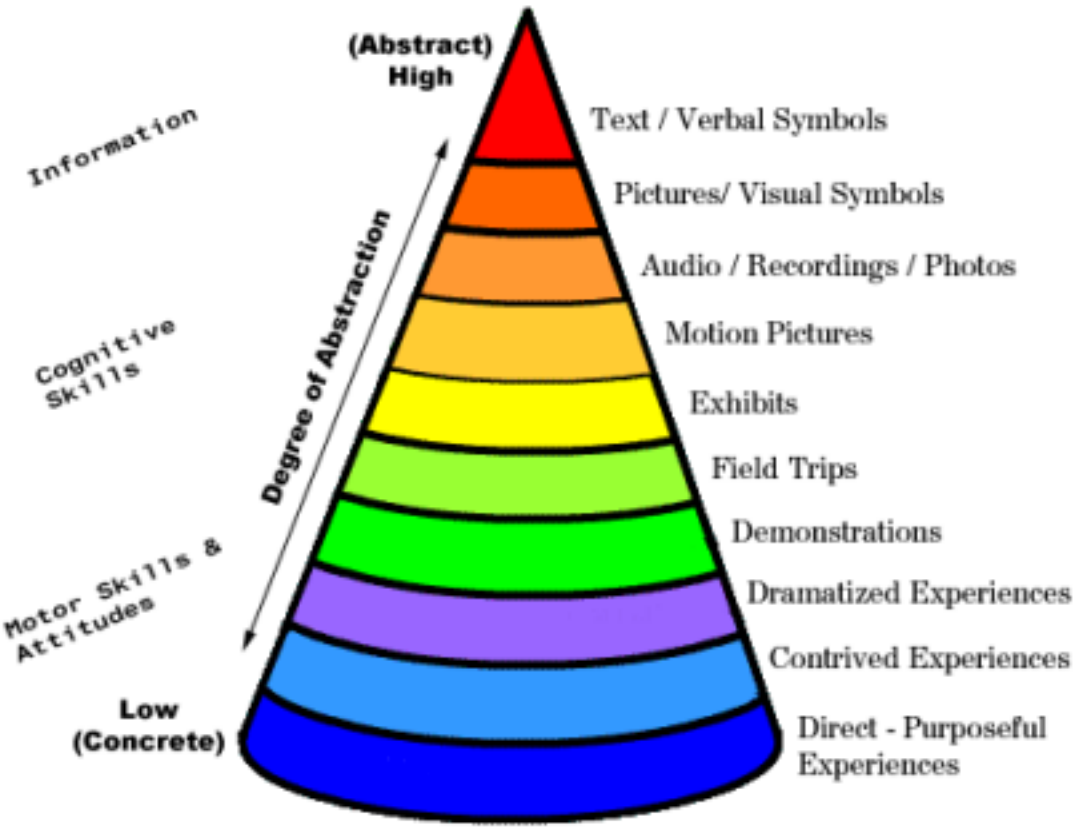
1. Familiarized with Dale's Cone of Experience and provided classroom processes or practices that exemplify each strata of the Cone of Experience
2. Provided examples of the various instructional materials appropriate for given instructional contexts.

Introduction

Edgar Dale has tried to provide a solid psychological base for the use of modern technologies through his famous "Cone of Experience". It is based on the type of experiences that as individual may acquire through the use of methods and equipment's related to hardware and software technologies. As shown in figure below Edgar Dale has arranged all the learning experiences from the point of view of the young learners in the order of increasing abstractness and decreasing concreteness.

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Technology for Teaching and Learning 1



Graphic courtesy of Edward L. Counts, Jr.

Principles on the cone of Experience:

✓ The cone is based on the relationships of various educational experiences to reality (real life), and the bottom level of the cone, "direct purposeful experiences," represents reality or the closest things to real, everyday life.

✓ The opportunity for a learner to use a variety or several senses (sight, smell, hearing, touching, movement) is considered in the cone.

✓ The more sensory channels possible in interacting with a resource, the better the chance that many students can learn from it.

✓ Each level of the cone above its base moves a learner a step further away from real- life experiences, so experiences focusing only on the use of verbal symbols are the furthest removed from real life.

1. **Direct Purposeful Experiences:** These experiences have least abstractness and maximum concreteness. The learner himself by coming into direct contact experiences them with the actual happenings n a real life situation. Consequently, we should always aim to provide real life practical experiences to the student in the shape of showing real objects and allowing him to come direct contact with the realities of the life through their own act.
2. **Contrived Experience.** These experiences are so planned and arranged as to provide a closest alternative to the real direct experiences in the situation where it is not possible or not feasible as well as practicable to have real life direct experiences. Contrived experiences may be provided with the help of some laboratory experiments or working models, miniatures or mock ups as a representation of reality.
3. **Dramatic Experiences.** This classification includes the experiences gained through their active participation and role playing in dramatic activities Experiences gained through such activities prove quite effective in the teaching learning of many subjects like history, political science, language and literature. These activities may have variety of form like full length play, one act play, puppet shows, pageants, pantomime, tableaux, dialogues on the spot spontaneous acting and mock experiences.
4. **Demonstration.** Here are the experiences are gained by the students through the observation of what is being demonstrated by the teacher on the demonstrations table in the form of actual objects, model of their working. For getting better results, a teacher should try to involve the students in the demonstration process through question/answers or taking their help in the arrangement and performances of demonstration.
5. **Field Trips.** These are the arranged situation in the form of educational tours, trip, excursion and outings may provide valuable opportunities to the students for giving real life direct experiences for the learning of many copies and gaining useful information knowledge and skills

(coupled with a lot of entertainment related to various issues of the school curriculum.

6. **Exhibition.** These are displays of models such as pictures, artifacts, posters, among others that provide the message or information. These are basically viewed, however, there are currently exhibits that allow the viewers to manipulate or interact with the display and as a result, the exhibit becomes more engaging and fun.
7. **Television and Motion Pictures.** These technology equipment provide a two-dimensional reconstruction of reality. These allow learners to experience the situation being communicated

through the mediated tools. They provide a feeling of realism as viewers try to understand the message portrayed by the actors.

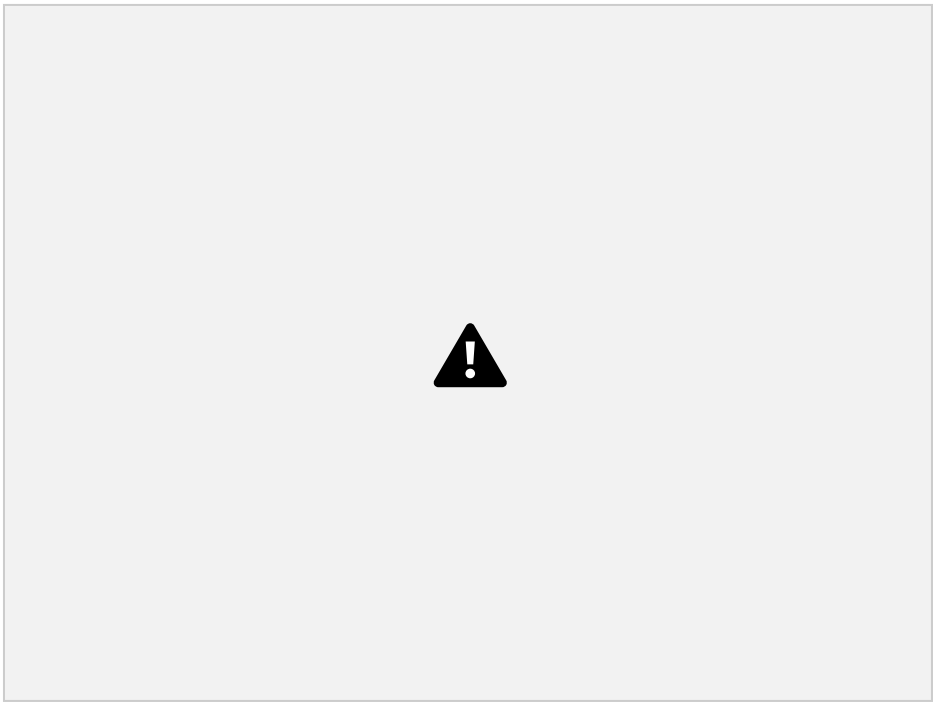
8. **Audio and Radio Recording.** Radio is an educational aid may provide valuable auditory experiences to the learners in terms of listening the specially arranged talks and discussion and information broadcast by the Radio Station in their schedule institution on general mass education programmed can be taped on audiotapes for timely references and use afterwards in classroom situation.
9. **Visual Symbols.** These are more abstract representations of the concept or the information. Examples of these are information presented through a graph or a chart. For example, a process can be presented using a flow chart
10. **Verbal Symbols.** These category appears to be the most abstract because they may not exactly look like the concept or object they represent but are symbols, words codes or formula.

In addition, Bruner's three-tiered model of learning points out that every area of knowledge can be presented and learned in three- tiered model of learning points out that every area of knowledge can be presented and learned in three distinct steps.

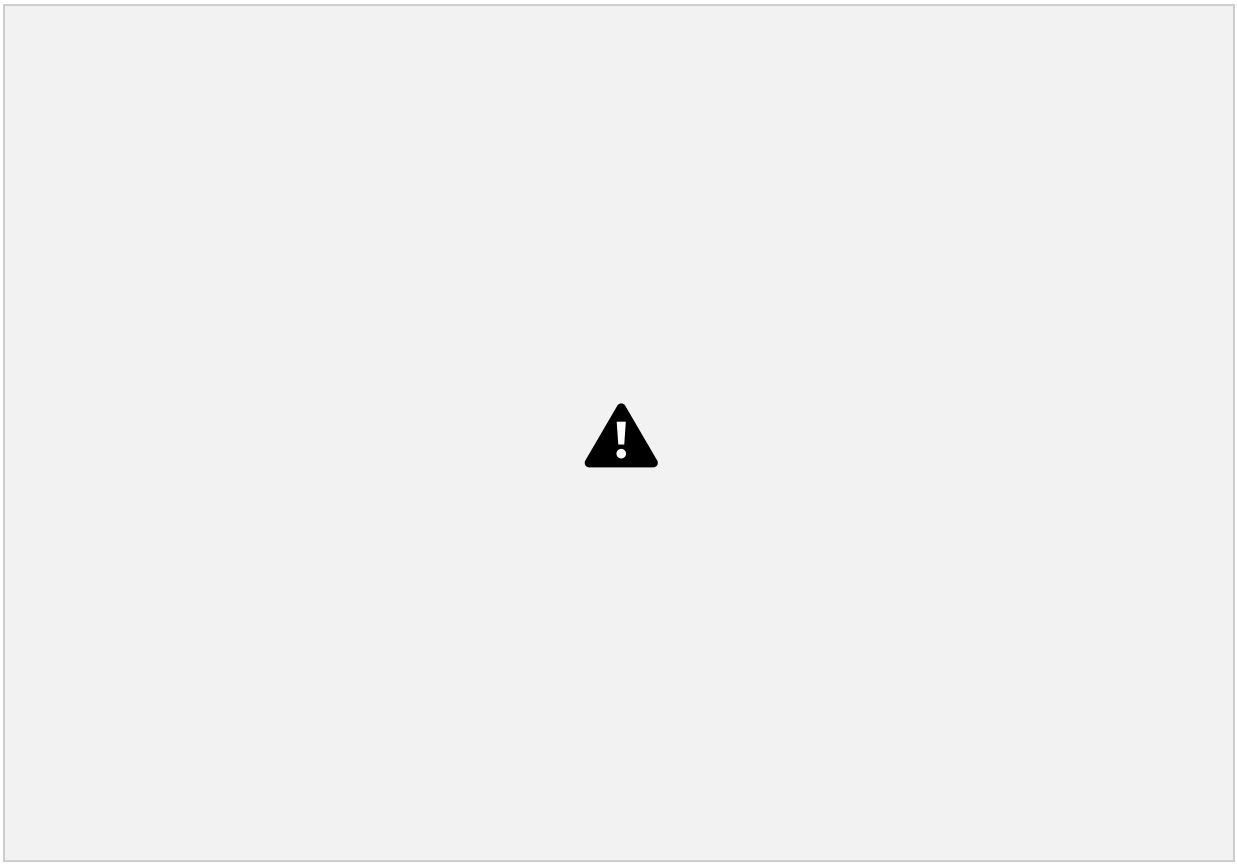
1. Enactive – a series of actions
2. Iconic – a series of illustrations or icons
3. Symbolic – a series of symbols

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Technology for Teaching and Learning 1



With young learners, it is highly recommended that a learner proceed from the ENACTIVE to ICONIC and lastly to the SYMBOLIC. A young learner would not be rushed to move to immediate abstraction at the highest level without the benefit of a gradual unfolding. However, when the learner is matured and capable to direct his own learning, it may move fluidly across the cone of experience.



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Technology for Teaching and Learning 1

To Do!

Learning Activity 4.1:Talk about your ideas given the following situations:

1. If you teach a lesson on the concept of fractions to a grader, how will \you proceed if you follow the pattern in Dale's Cone of Experience beginning with the concrete moving toward the abstract?
2. Try to explain why reading teachers discourage us from reading only comics or illustrated comic version of novels?
3. How would you account for children who can label and identify the objects even if they have not actually seen them in reality?
4. Now that there is a great deal of ICT tools and applications used in enriching the lesson, how would you explain its potential use in instruction and where will you categorize it on the Cone?

Reflective Essay 4. 1: In writing your reflection always use Gibb's Reflective Model and discuss the

concept below:

Dale's Cone of Experience is a tool to help instructors make decisions about resources and activities. As you prepare to become a teacher, you can use the concepts of Dale's Cone of Experience and ask the following questions:

1. Where will the student's experience with this instructional resource fir in the cone?
2. What kind of learning experience will you choose for your students?
3. How will you use the ideas in the cone to enrich your textbook?
4. What instructional material (digital or non-digital) will you use to enrich your students' learning experiences?
5. How many senses will your students employ when you use an instructional material taken from a band of the cone?

LESSON 2

TPACK as a Framework for Technology-Driven Teaching and Learning

Lesson Outcomes

1. Familiarized oneself with TPACK and SAMR Model as framework integrating technology in the practice of teaching
2. Selected and employed as appropriate technology (tool or application) in an instructional plan
3. Evaluated one's capability in integrating in his lesson
4. Identified learning theories and principles applied in the design and development of lessons through appropriate media and technologies for teaching-learning

Introduction

As you prepare to be a teacher, how do you assess your content knowledge of your specialization? In terms of your teaching skills, what strategies or techniques do you know will work if you use it when teaching? If you will consider using a technological tool when teaching, what would it be?

Shulman's (2008) Pedagogical Content Knowledge (PCK) is a framework which involves the teacher's competence in delivering concepts being taught by simplifying complex ones if needed or leading the students to study a concept more deeply and extensively. This is made possible due to the teacher's understanding of the amount and the content structure of knowledge.

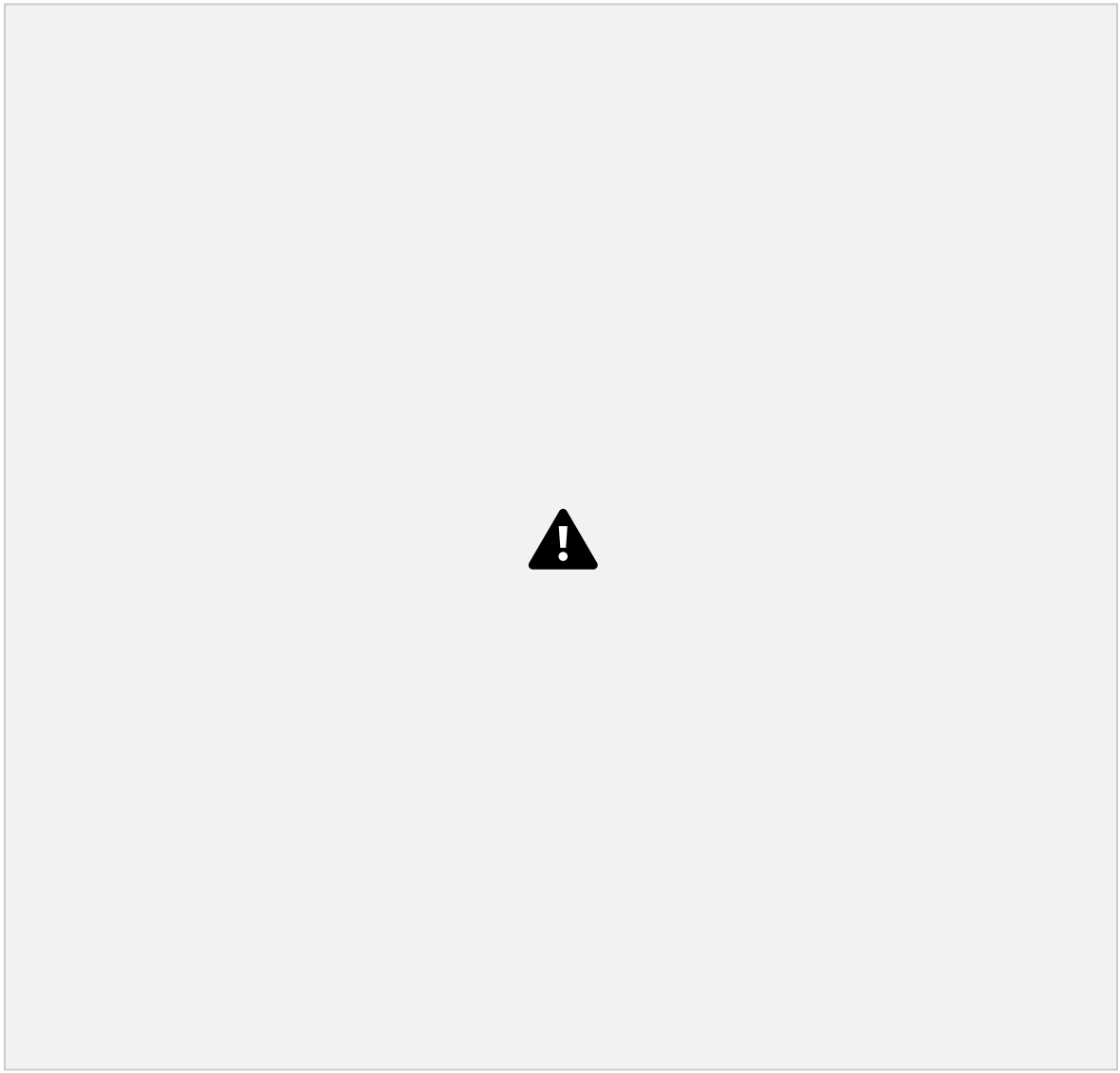
Mishra and Koehler (2006) continued to build upon Schulman's PCK and incorporated technology hence, TPACK.

Discussion

What is **TPACK?**

TPACK is a framework that combines the teacher's knowledge areas: **technological knowledge, content knowledge** and **pedagogical knowledge.** This framework shows the interconnectedness of content knowledge more engaging, relevant and effective. It looks at how these knowledge areas intercept or work together to increase student's motivation and make the content

more accessibly engaging to students. This framework has significantly influenced practices in teaching as a result of research.



The Technological ,Pedagogical and Content Knowledge known as TPACK is a theoretical model describing the capability of the 21st century teacher. To be relevant to the learners of today, the teachers should be able to blend his knowledge and capability in the content of the discipline.

CK is **the content knowledge**. It is 'what' – understanding of the content such as language, arts, math, science, history among others. This is composed of facts . concepts, theories or principles in a given discipline. **Pedagogical Knowledge (PK)** is the 'How'. It is the expert's knowledge in the science of teaching from educational and learning theories to individual differences to strategies and

techniques as well as assessment of learning. The teachers need to possess the capability to handle

learning through effective method and appropriate strategies.

The intersection of the pedagogical and content areas of knowledge is the **PCK.** This is how much competencies the teachers have in making the students learn concepts and skills. This includes the knowledge of how to use techniques that can meaningfully address different learning styles while supporting content with deeper understanding. Schulman (2008) considers this teaching at its best.

The incorporation of technology knowledge into PCK by Mishra and Koehler (2006) is relevant to 21st century teaching. This is the **TK or the technological knowledge**. This is the teacher's knowledge on how to select, use and integrate these tools in the teaching and learning context. This is not only about the tools but also about the quality of content that students can access through appropriate applications and sites.

When technology is used in a specific subject area to enrich and deepen student's understanding of content, this is the intersection called **TCK or Technological-Content Knowledge.** For example, to deepen the student's understanding of a concept or area of study, students can gather information and collect data and evidence using technology tools and present these information using an application.

Furthermore, **TPK** which is **Technology-Pedagogical Knowledge** knows how to select, use or develop technology to manage student learning. For example, who will use collaborative tools that will allow students to work together and share information? How will you use an application that will allow students to creatively share their well-researched project in class or to others outside of the school? If so that they may be able to interview the right expert who can assist them in their capstone project? As a teacher, using the appropriate technology to ensure effective instruction is TPK.

It allows student to learn content through digital tools. They can document and record evidence and plot out strategy using a collaborative tool where the group can discuss their action plan while in or out of school. The drawn plan can be presented using a productivity tool (i.e. PowerPoint presentation or a digital story) to pertinent persons for feedback. Once approved and given support, the project can now proceed to the next-implementation stage.

The intersection of all knowledge areas is the **TPACK**. So, TPACK, is about what teachers know, how they teach and how technology is used in the delivery of the lesson to make it more engaging and making learning more relevant in the 21st century.

In conclusion, TPACK will start with content and then pedagogy and layer in technology. Sometimes, you might get excited about the technological tool and design a lesson around it. Remember, it is NOT about technology tool but it is the right blend of the three knowledge areas.

To Do!

Learning Activity 4.2.A: To elaborate about TPACK**,** watch the videos from Youtube and provided below are the topic and its link then answer the questions below.

a. TPACK in Two Minutes https://www.youtube.com/watch?v=FagVSQlZELY

b. Introduction To Tpack Model Video Common Sense Media https://www.youtube.com/watch?v=glkn9Veggxo

c. TPACK https://www.youtube.com/watch?v=0AiUNujA0DM

Questions:

1. Describe the relationship of pedagogical knowledge with content knowledge as illustrated from the video.

2. Describe the relationship of the technological knowledge with content knowledge. 3. Describe the relationship of the technological knowledge with pedagogical knowledge. 4. As pre-service teachers, how will this TPACK integration model guide you in using technology in the delivery of instruction?

Reflective Essay 4.2.A: In writing your reflection always use Gibb's Reflective Model and discuss the concept below:

Evaluate your personal capability in relation to the TPACK framework. How does your TPACK look like?

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Technology for Teaching and Learning 1

SAMR

Another important model in the integration of technology in instruction is SAMR.

SAMR is a **model** designed to help educators infuse technology into teaching and learning. Developed by Dr. Ruben Puentedura, the **model** supports and enables teachers to design, develop, and infuse digital learning experiences that utilize technology. (Nov 9, 2013)

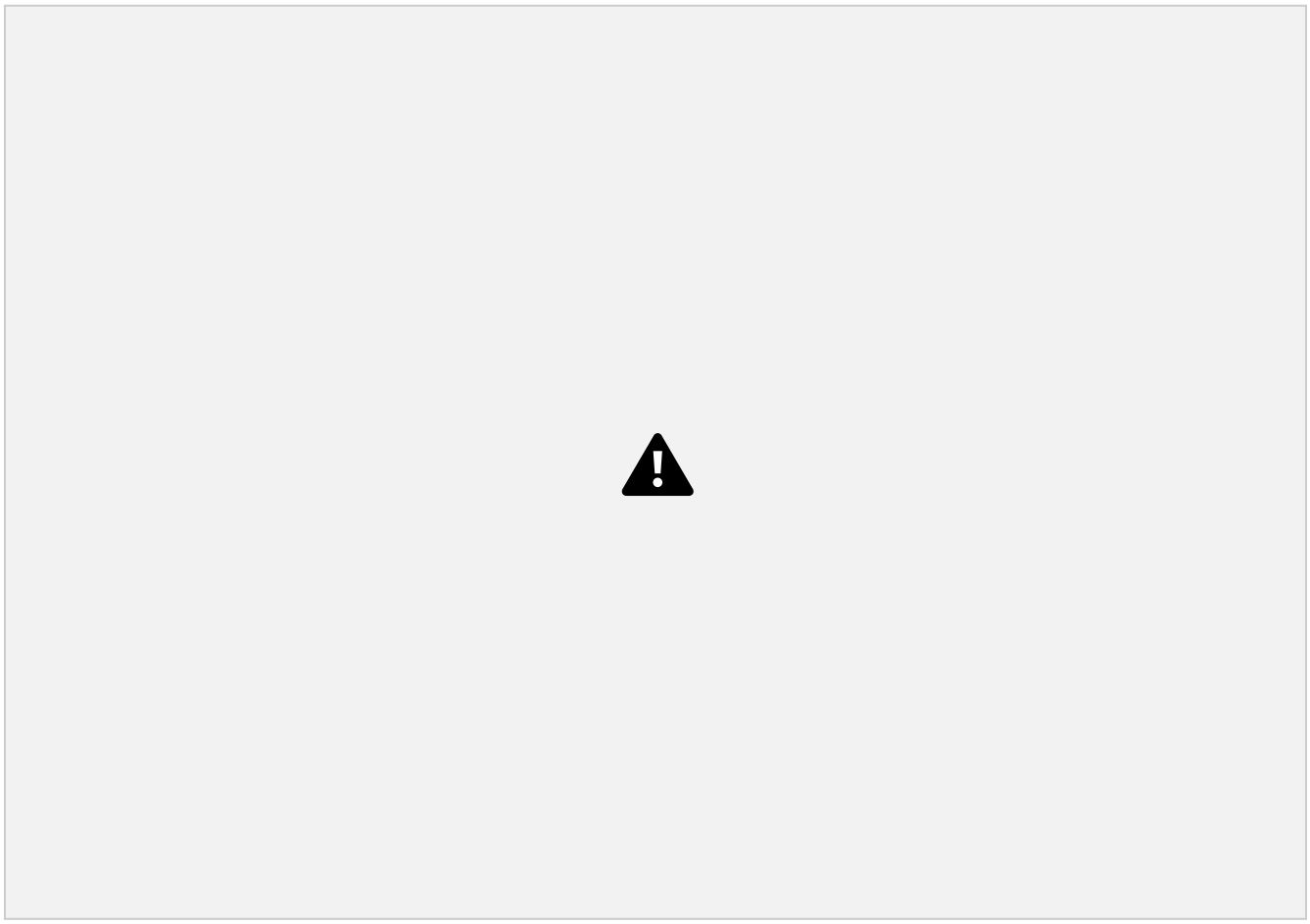
S stands for SUBSTITUTION. It is literally using technology as an alternate for the regular item that is employed in augmenting or assisting instruction. Instead of using a map or globe in teaching geography, teachers can use Google map or Google earth instead. Showing a location in a map can be substituted by the use of google map or google earth.

A is AUGMENTATION. In this part of technology integration, there is a functional improvement in the use of technology tool. With the google map, map you can use the features available such as measuring the distance from one point to another. You can provide a task by making students utilizes this. For example, ask them to measure the distance and get the estimate time of travel if they start from one point to go to the next point. With this use of the google map, you tap on certain functions that the application provides. Instruction with technology integrated in enhanced.

M means MODIFICATION. In this level, technology is used to redesign a lesson. When studying geography for example, google applications have panoramio. Using this you can guide students to start uploading their own pictures of places they may have visited online. A brief description of these places can also be added, If internet connectivity is not available in the classroom, they can do the task after class. With clear and proper instructions, students can be guided to work on the assignment. **R** represents the idea of creating something from the regular item. This is REDEFINIION. Technology allows the creation of a new concept that has not been previously conceptualized. Puentedura labels this as the transformation of learning. Students are given the opportunity to create an output based on what they know and learned using technology tool or application. So, with the ready pictures uploaded in panoramio, students can probably write the script for a guided tour using his available materials (i.e.

pictures) using Google Earth and start arranging these to be shared through the world wide web. It would be good if before the students start posting or publishing their work online, as a teacher, you can review it just to be sure it is ready to be shared.

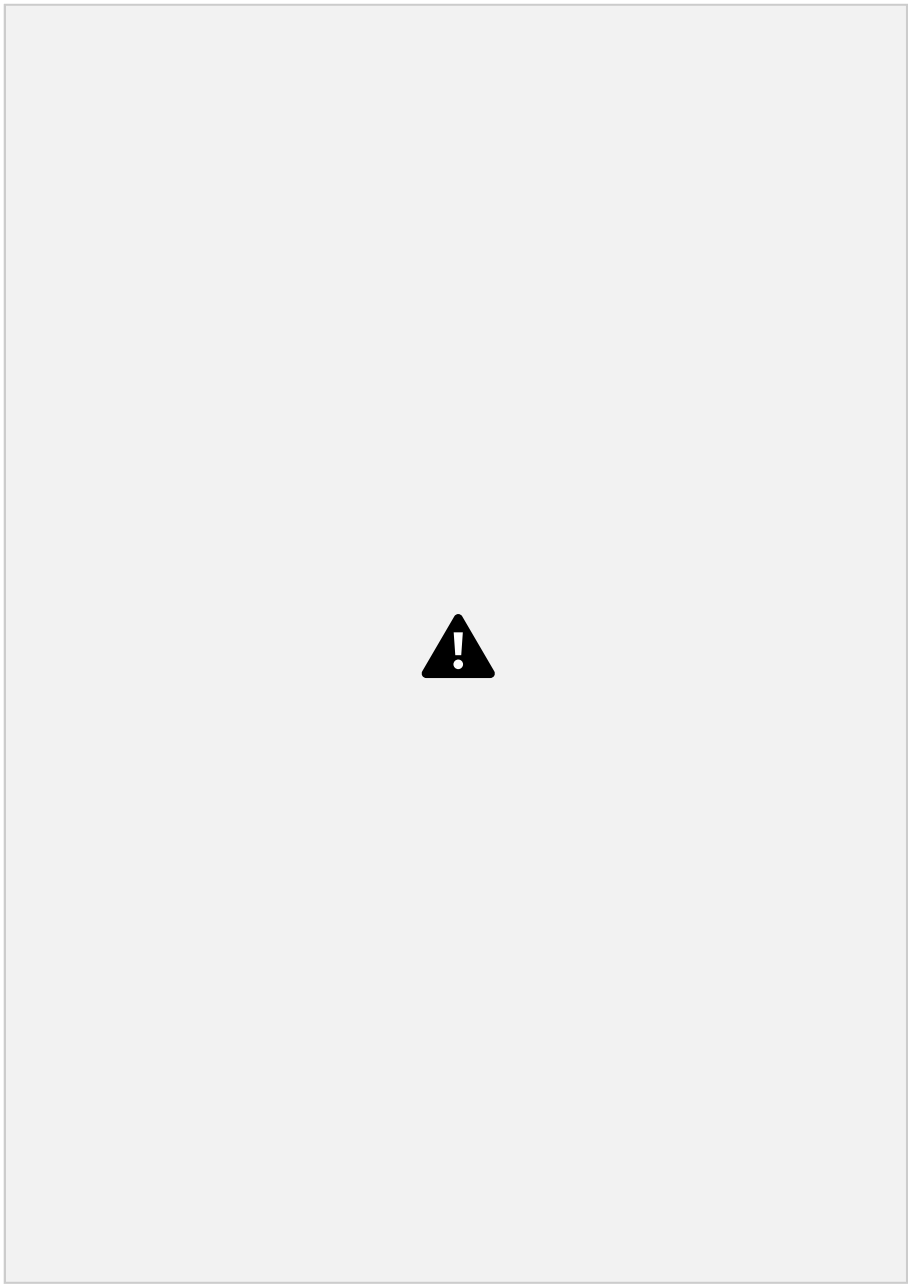
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The figure above provides examples of integrating technology in some learning areas.

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Technology for Teaching and Learning 1

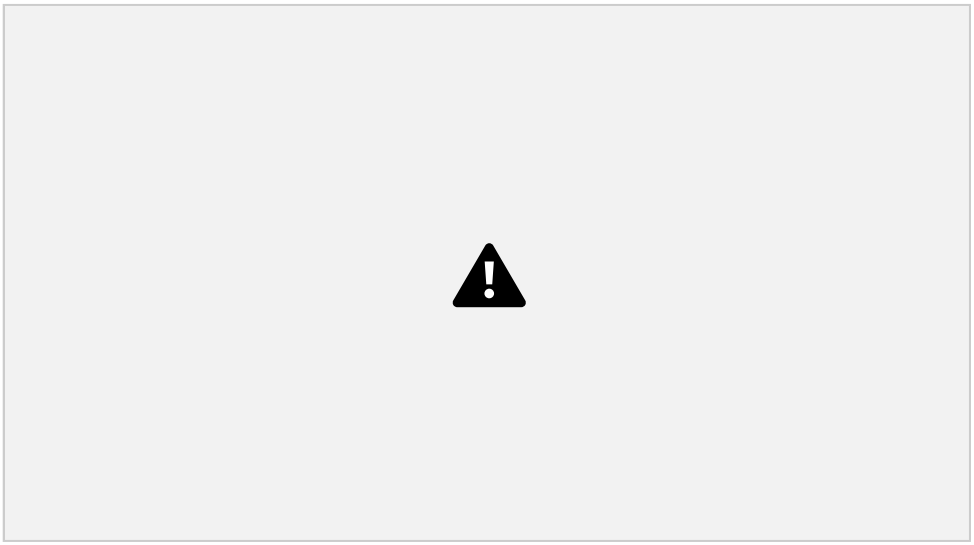


This may help….

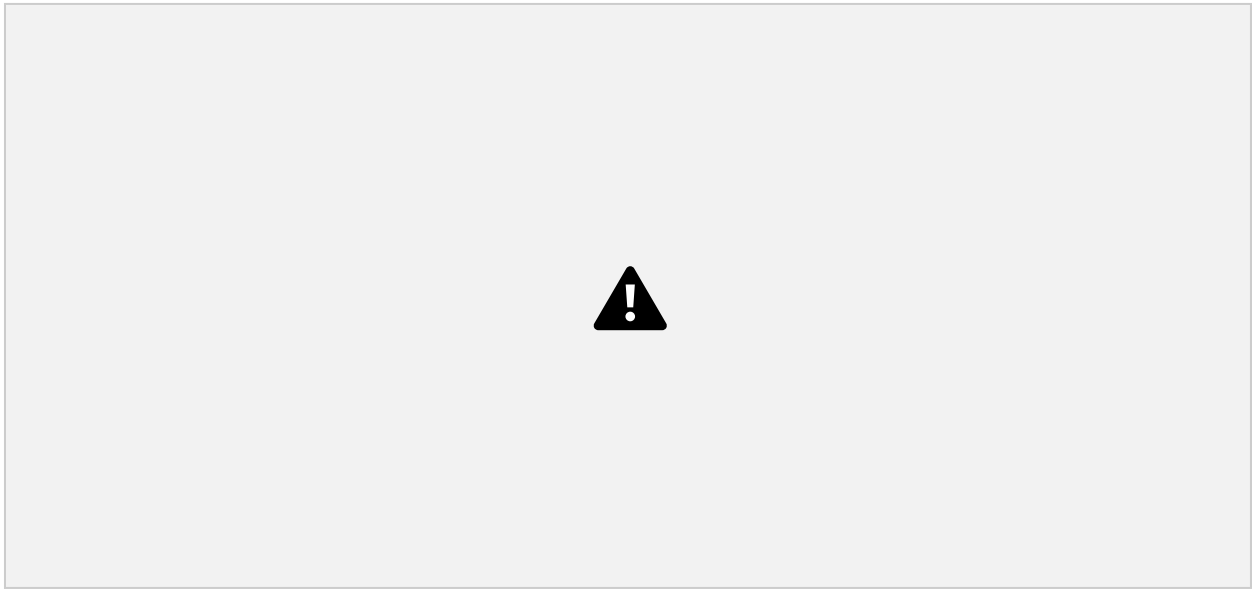
You may use applications that you know. If you have no idea, you may start exploring some tools and application found in the image of Apps in Education Poster and think how you can integrate them in the lesson. All you need to do is to explore them and be familiar with the features.

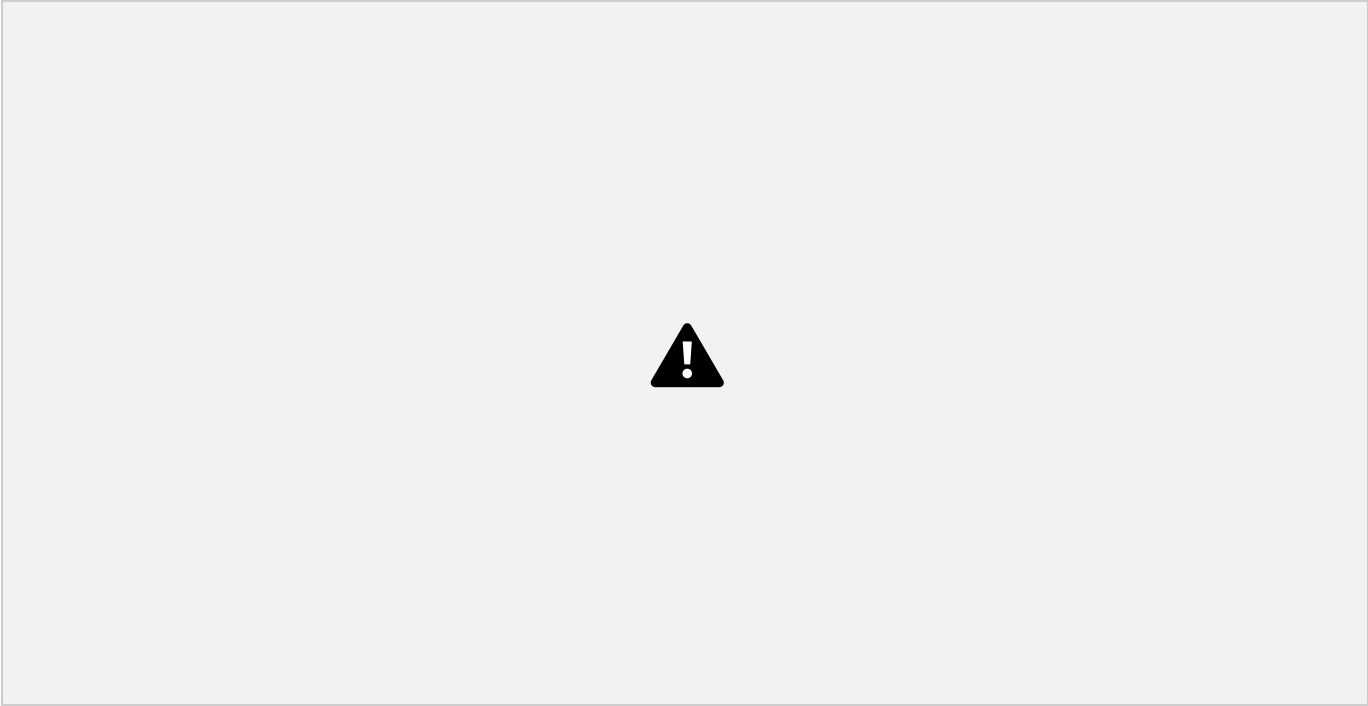
Blooms' Digital Taxonomy

Benjamin Bloom is a known educationist who provided the cognitive taxonomy which has greatly been used in education. His students, Anderson and Krathwoll made some revisions. This is the Revised Bloom's Taxonomy



This figure shows the comparison between the Original and the Revised Bloom's Taxonomy





The infographic above shows the relation of SAMR and Revised Bloom's Taxonomy

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Technology for Teaching and Learning 1

To Do!

Learning Activity 4.2.B: Using SAMR, think of how youy can integrate a technology tool for each phase to show the transition from using it to enhance instruction leading to a transformative use of technology. Design a task that can have a positive effect on student's learning outcomes.

	Task or Learning activities
Redefinition	
Modification	
Augmentation	
Substitution	

Reflective Essay 4.2.B: In writing your reflection always use Gibb's Reflective Model and discuss the concept below:

What is the significance of being aware of a theoretical framework (i.e. TPACK, SAMR, Bloom's Revised Taxonomy) in preparing to become an effective teacher?

Lesson 3

The ASSURE Model

Lesson Outcomes

1. Identified learning theories and principles applied in the use and design of learning lessons with technology
2. Applied the ASSURE Model in crafting a lesson.

Introduction

A well-planned lesson is half of the battle won; the other half is the delivery of the lesson by the facilitator who has the pedagogical skills in implementing what has been planned. This will ensure the achievement of learning outcomes. This will engage learners to perform expected standards. That is

why, it is important that before teaching, elements needed to be incorporated when designing a plan have to be plotted well.

Discussion

The ASSURE Model

ASSURE Model is a guide in crafting an instructional flow that can guide the teacher in integrating appropriate technology and media into the instructional process. It is also refers to a systematic approach that the teacher can use when writing an instructional plan. A new model to help assure learning, developed by Heinich, Molenda and Russell (1982), is the ASSURE model. The acronym stands for the following:

A = Analyze learner characteristics

S = State objectives

S = Select, modify or design

U = Utilize materials

R = Require Learner Response

E = Evaluate

Analyze Learner Characteristics

The first step in planning is to identify the learners. Learners may be students, trainees or members or organization. A teacher must know his students for him to be able to select the "best" medium to meet the objectives. The audience can be identified by two types of characteristics: a) general characteristics; and 2) specific entry characteristics such as knowledge, skills and attitudes about the topic.

State Objectives

The objectives may be derived from a course syllabus, stated in a textbook, taken from a curriculum guide or developed by the instructor. It should be stated in terms of what the students will be able to do as a result of instruction. The conditions under which the learner is going to perform and a statement of acceptable performance level should be included.

Select, Modify or Design

The task is to build a bridge between identification of audience and objectives are stated. These are two options: a) select available materials; b) modify existing materials; or c) design new materials.

Utilize Materials

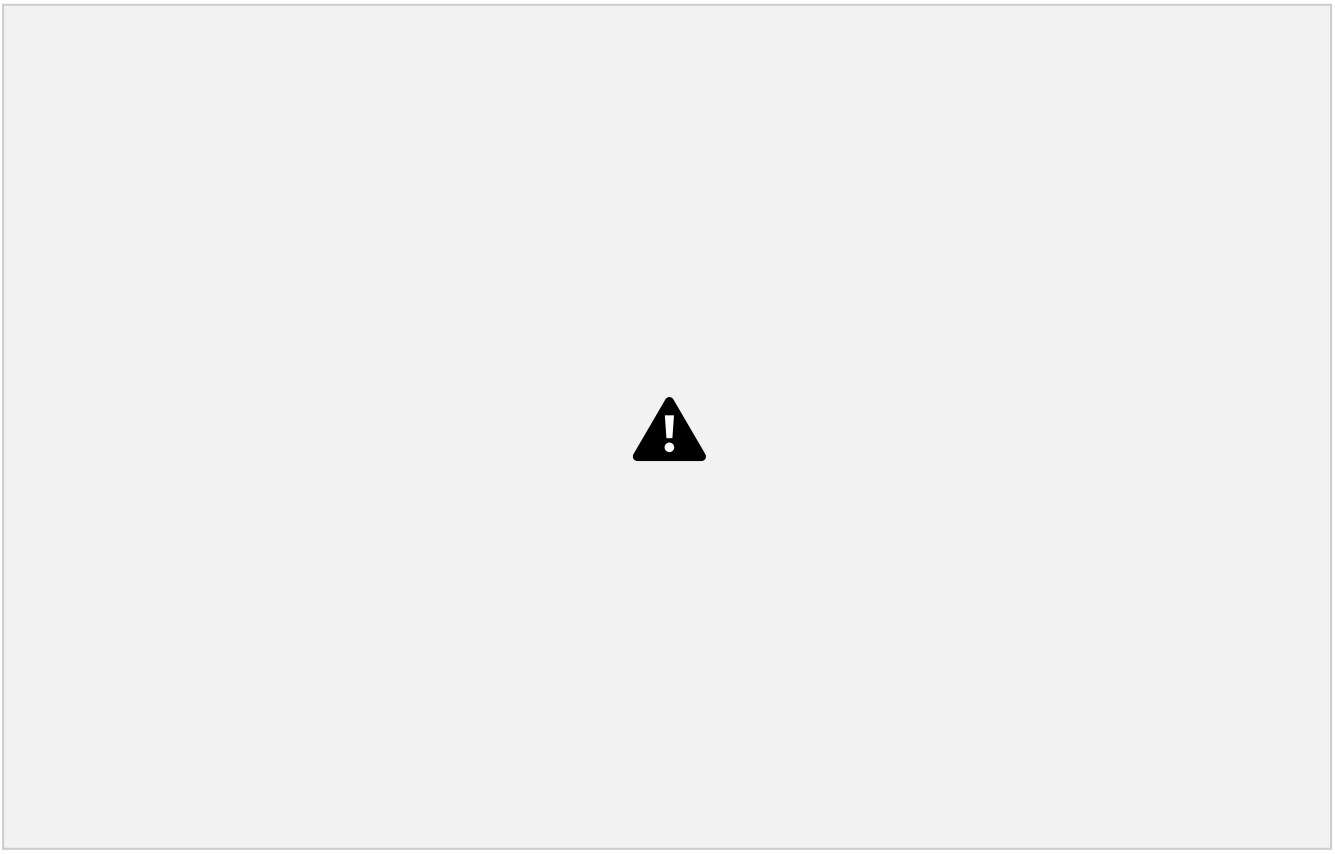
After the materials have been selected, modified or designed, plan how these will be used and how much time will be spent using them. Next, prepare the class and ready the necessary equipment and facilities. Finally follow up with class discussion, small group activities or individual projects or reports.

Require Learner Response

To assure learning, students must practice what they are expected to learn and should be reinforced for the correct response. The first time that student is expected to perform the behavior called for in the objectives should not be on the examination. Instead, there should be activities within the lesson that allow students to respond and to receive feedback on the appropriateness of their performance or responses.

Evaluate

After instruction it is necessary to evaluate its impact and effectiveness. To get the total picture, evaluate the entire instructional process. Did the students meet the objectives? Did the media assist the student in reaching the objectives? Could all students use the materials properly? Was the environment comfortable? Did the teacher facilitate the student learning by providing the necessary assistance for the individual students?



To Do!

Learning Activity 4.3: Select any lesson in your area of specialization (ex. Elementary Education) and a think of a hypothetical group of learners. Design the lesson good for an hour by writing a brief outline of the plan. Explain how the ASSURE Model has helped in the process of designing it. Discuss the rationale of the decision on the selection of the materials, technology tools, learning and assessment activities.

Note: I already fill in the table with my example.

Elements in an Instructional Design	**What is it?**	**Explain the reason for the decision in relation to the ASSURE model.**
Lesson Topic:	Ex. Describing the effects of wind, given a certain storm warning signal	(It's up to you to fill discuss here and the succeeding columns)
Subject Area:	Science	

Level of Learners:	Grade 5 Average pupils	
Duration:	One hour lesson	
Learning Outcomes:	• Describe the effects of the winds, given a certain storm warning signal • Act on a role playing on preparing ourselves for the incoming typhoon. • Appreciate the importance of heeding or listening to weather news	
Materials/Technology Tools:	Downloaded video, TV pictures,	
Learning activities:	1. Picture Recognition. The teacher will show pictures of the aftermath of a typhoon in the different affected provinces in the Philippines 2. Role Play. The pupils will dramatize the on the preparation of typhoon by news broadcasting.	
Assessment Activities: Pen and paper test	The pupils will be provided with a chart to identify the speed of the wind and its impact on a given storm signal number.	

Reflective Essay 4.3: In writing your reflection always use Gibb's Reflective Model and write your reflections gained from the lesson. The succeeding questions can guide you in the process.

- What have you learned from this lesson on ASSURE model?
- How will you use what have you learned from the lesson?
- What did you find difficult to understand?
- If you find an item or a step difficult, what action will you take to address your need?

Module 5

Innovative Technologies for Assessment Tasks in Teaching and Learning

Module Outcomes

1. Explained the role of ICT in assessment in learning
2. Described the current trends in technology-based assessment practices in schools
3. Selected appropriate assessment tools

Introduction

Assessment, learning and teaching are closely connected and interwoven because each provides information that can be effective instruction. In the process of conceptualizing a plan on how to teach, as the learning outcomes are put in place, the next step to consider is how these are to be evaluated. This is an important part of instruction because data gathered in this phase of instruction will inform the teacher whether the students have actually learned or that the teacher has to reconsider a better approach in teaching.

Although the conventional ways of assessment still work, the role that ICT plays in the present teaching and learning context has definitely changed the course of direction that a teacher can pursue. The use of technology plays a vital role in assisting teachers how to assess students' learning. It involves digital devices or online applications that can be used in constructing, delivering and reporting learners' performance.

Lesson 1

ICT and Assessment of Learning

Lesson Outcomes

1. Explained the concepts of the role of ICT in assessment

Introduction

Assessment is very important in student learning. There are many approaches to assessment in learning with an emphasis on the process of learning along with the product of learning. The ultimate measure of student's competence is performance. Development in Information and Communication

Technology (ICT) has thrown open numerous possibilities for assessing student learning and providing immediate feedback.

Discussion

Assessment is the process of identifying, gathering and interpreting information about students' learning. To understand better the different related terms on assessment especially formative and summative assessments, please watch this video of YouTube.

Types of Assessment

1. Formative Assessment – provides feedback and information during the instructional process, while learning is taking place, and while learning is occurring.

2. Summative Assessment - takes place after the learning has been completed and provides information and feedback that sums up the teaching and learning process.

3. **Diagnostic Assessment** – can help you identify your students' current knowledge of a subject, their skill sets and capabilities and to clarify misconceptions before teaching takes place. Knowing students' strengths and weaknesses can help you better plan what to teach and how to teach it.

4. **Authentic Assessment** – describes the multiple forms of assessment that reflect student learning, achievement, motivation and attitudes on instructionally relevant classroom activities.

The Role of ICT in Assessment in Learning

There are new technologies created to provide students with higher level thinking skills, particularly the aspects of creative and critical thinking, and the opportunity to teach and assess those skills. The use of ICT in assessment is very helpful to teachers because students' information and results of examination can be recorded and stored and can easily be retrieved. With the use of ICT, the

teacher can right away give g=feedback to students on the results of assessment. Feedback from the computer during the use of test materials improves student performance in later use of the same test material.

Tests play a vital role in traditional learning as well as e-learning and this helps the teachers and learners in measuring learning. There are many freeware where the teachers can create quizzes and evaluate learning electronically. Online examination systems seek to efficiently evaluate the test

takers thoroughly through a fully automated system that not only saves time but also gives fast results. It is done in most case through a Web-Based Online Examination Software. This online examination will also eliminate the need for monitoring while the exam is being taken.

One of the advantages of using a web-based exam software or an online examination system is that it gives a high level of transparency as opposed to the traditional method or remote method. It is almost impossible to compromise exam questions and evaluations because they cannot also be influenced. Most online exams generate their results immediately. Some of the major advantages of online examination are: (1) accessibility and flexibility; (2) time management; (3) save cost; and (4) statistical analysis.

Security and confidentiality should also be considered. Security needs to be given top priority for any website that you will add information to. Some schools utilized software for security and privacy of examinations.

Current Trends in Assessment in Learning

Traditional assessment like objective test such as multiple choices, matching type true or false and other paper-and-pencil test received criticism by some experts in assessment. The 21st century assessment includes: (1) utilizing multiple measures of assessment. (2) Increasing the use of technology, (3) focus on the growth and development of the learner, and (4) differentiating the roles and responsibilities of the teachers, administrators and other stakeholders. To help students become college and career-ready, we need to teach them how to apply what they are learning in school to practical and intellectual tasks in their everyday lives (Wolpet-Gawron, 2010).s

The concept of 'classroom' has been shifting in recent years with dedication to purposefully designed learning spaces and seamless integrations of technologies (Williams, 2017). In the 21st

century, we are facing a global economy where information travels at the speed of light and knowledge of how to harness and sift through that information has become vital to our personal and national well being. As an educator, we know that our students must graduate ready to function in this expanded world. Educational technology is used to increase the efficiency of education in teaching. Learning and assessment. Computers and related technology are viewed as the future of teaching and learning and also as powerful technological machine to promote developing and learning. Computers are able to create a more attractive and effective learning environment (Keser &Ozcan, 2011).

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Education technology has introduced new, exciting and innovative ways to engage students in active learning while online. The teachers can incorporate active learning activities through technology in fully-online lessons, homework modules, and even in-class online activities. Roblyer and Edwards (2000) as cited in (Keser & Ozcan, 2011) suggested that there are important reasons for teachers to use technology in education: motivation, distinctive instructional abilities, higher productivity of teachers, essential skills of information age, and support for new teaching techniques (cited in Samak, 2006).

Online learning initiatives often fail to engage and tech learners because they are passive in nature – reading PDFs, flipping through slides, listening to long lectures. One way to improve these experiences is to focus on incorporating active learning methodologies.

Active learning is any learning activity in which the student interacts or engages with the material, as opposed to passively taking in the information. The parents can also help the learners engaged in rich learning experiences at home and during daily activities such as field trip.

The teacher will give learning activities that are intentional, meaningful and useful. Activities which involve student interaction with content can include listening to and/or watching a live or recorded talk, engaging with a written or visual text, engaging with multimedia, or a combination of these.

To Do!

Learning Activity 5.1: Conduct an interview with your teachers regarding their practice on assessment. Make an unstructured interview guide.

Reflective Essay 5.1: In writing your reflection always use Gibb's Reflective Model and reflect in answering following questions:

1. How do students know what is expected and are they are able to articulate expectations?
2. What is the role of ICT in assessment of learning?

Lesson 2

Criteria in Choosing Appropriate Assessment Tools

Lesson Outcomes

1. Examine appropriate assessment tools to be used in the classroom
2. Reviewed assessment used by the teachers and determine if it is applicable in the 21st century

Introduction

Assessment methods are the strategies, techniques, tools and instruments for collecting information to determine the extent to which students demonstrate desired learning outcomes. Teachers need to select tools to assess student achievement in the target outcome(s). The selection of tools involves a process to obtain detailed information and the need to keep the process feasible and manageable. Student learning styles vary widely and their strengths and challenges with the respect to assessment vary as well. Teachers need consider that variation as they choose assessments for their courses. By varying the way teachers assess student understanding, they are more likely to offer opportunities for every student to demonstrate their knowledge. The first step in attaining competency in selecting appropriate test involves understanding the purpose or purposes for which an assessment is given. According to Mehrens (2001) as cited in McDivitt and Gibson (n.d.) in its broadest sense, the purpose of any assessment is to gather data to facilitate decision making. However, many kinds of decisions and many different types of information may be gained from the use of tests and may serve to

facilitate decision making. The following may be considered in choosing appropriate assessment tools; (1) goals of assessment (2) methods of assessment, either direct or indirect; and (3) the interval in giving assessment.

Discussion

One of the challenges for beginning teachers is to select and use appropriate assessment techniques suited to their learners' needs. During teaching, teachers not only have to communicate the information they planned but also continuously monitor students' learning and motivations have to be made (Airasian,2005). Beginning teachers find this more difficult than experienced teachers because of

the complex cognitive skills required to improvised and be responsive to students' needs while simultaneously keeping in mind the goals and plans of the lesson (Borko & Lvingston,1989). The informal assessment strategies teachers most often use during instruction are observation and questioning.

The teacher may either be utilizing direct or indirect method. Examples of direct methods are examinations, written assignment, oral presentations and performances, internship supervisor's ratings of student skills, portfolios, score gains between entry and exit tests, capstone projects, theses and dissertations, etc.

Indirect methods can provide a useful supplement and check on the findings from direct measures. The following are examples of this indirect method: Student satisfaction surveys, surveys of students and alumni, exit interviews with graduating students, student participation rates, reflective essays, data on placement and other measures of post-graduation performance, etc.

The teacher may consider the following characteristics in selecting appropriate assessment tools whether conventional or ICT-based assessment:

1. Measure the desired level of performance (level of satisfaction, productivity, efficiency, student performance)
2. Cost effective in terms of effort, time and money
3. Useful that will produce results that provide information that can be used in making decisions to

improve student learning

4. Reasonably accurate and truthful
5. Dependable, consistent responses over time
6. Evidence of being on-going, not once and done

After the discussion of a certain lesson the teacher conducts formative assessment. One of the key characteristics of formative assessment is the frequent use of evidence gathered from various methods as feedback. Besides serving as an indicator of attainment levels, students use this feedback to set goals for the next stage of learning. With this, teacher may utilize ICT in assessment. It enables both teachers and students to provide valuable feedback on each learner's progress. The rapid development of ICT has provided teachers with digital platforms that support learning and teaching (Woo et.al., n.d.). Online task assigned by teachers will assess the collaborative problem solving construct and the five strands (Participation Perspective Taking, Social Regulation, Task Regulation as well as students'

Learning and Knowledge Building skills) according Drigas and Karyotaki (2006). It is also stressed in the same study that large-scale differential item functioning (DIF) analysis can validate an online problem solving ability test that minimizes the extraneous differential effects of students' language background. Thus, online assessment can be incorporated by teachers. ICTs provide the means for the explicit design and implementation of individualized or collaborative problem-solving tasks through teachers' interactive scaffolding and immediate feedback supply towards learners (Drigas & Karyotaki, 2006).

Creating and Utilizing Rubric

A rubric is a set of criteria used to determine scoring for an assignment, performance or product. This can be used to score many kinds of written assignments or exams, papers, project, speeches or eProtfolios. They are no useful, however, as a grading mechanism for multiple choice or short answer tests. There are many samples of rubrics online, but the teacher may consider the learners, objective of the lesson, contextualization and localization. The main purpose of rubrics is to assess student performances. For some performances, the teacher may observe the student in the process of doing something, like making their projects, online drill, tutorial, and many more. There are two (2) types of rubric:

1. Analytic rubrics describe work on each criterion separately. It utilizes separate, holistic

ratings of specific s=characteristics, products, or behaviors.

2. Holistic rubrics describe the work by applying all the criteria at the same time and enabling an overall judgment about the quality of the work. It utilizes holistic rating for a product or behavior.

Technology can be used for assessing student learning in various purposes. By using technology in managing assessment information data can be presented in different ways to meet the needs of the students, teachers and administrators. An electronic examination which is also called computer-based assessment (CBA), computer-based testing (CBT) or e-exam is a test conducted using a personal computer (PC) or an equivalent device, in which the delivery, responses and assessment are effected electronically. e-exams were developed more than four decades ago for professional certification in the IT industry and progressively evolving as a preferred alternative to Paper-Pencil-Test (PPT) in schools, universities, recruiting firms as well as private and public organizations (Bulama Bukar,Bello & Baba Ibi, M, 2016).Computer-Based-Testing is more efficient that Paper-based test because it is an individualized

testing and has faster score reporting within few minutes after last submission. The students can immediately view the scores on screen and more convenient for teachers and students.

Computerized delivery of objective tests has more advantages compared to paper-pencil-test which include the following:

1. The creation of item bank of questions invites the possibility of each student being presented with a paper made up of different questions, but of an equivalent standard. 2. Automatic computerized marking facilities as they find useful.
3. Students can be invited to sit as frequently as they find useful.
4. Computerized recording of results facilitates the analysis of groups' responses to questions.

The leader may consider balance (Blend of traditional and ICT based assessment). Although computerized testing facilities can provide a rapid means of assessing and providing feedback to large numbers of students, it is essential to consider their use as part of the overall unit strategy, especially as multiple choice/limited response type questions can lead to an emphasis on "shallow" learning. Electronic assessment tools are unlikely to reduce significantly the burden of assessment, but they can be used to promote deeper and more effective learning, by testing a range of skills, knowledge and understanding. Using computers in assessment does not have to mean more multiple choice testing to

the exclusion of other assessment techniques. A wide range of innovative assessment methods lend themselves to computer-based implementation.

To Do!

Learning Activity 5.2: Research on some online assessment tools and explain how it works.

Reflective Essay 5.2: In writing your reflection always use Gibb's Reflective Model and discuss the answer on the following questions:

1. What is the role of ICT in assessing student learning? Cite specific examples?
2. As a future teacher, how will you assess your future students?
3. What are the advantages and disadvantages of using ICT in assessment? How about conventional assessment methods?
4. Cite ways of selecting appropriate assessment tools.

Module 6

Social, Ethical and Legal Responsibilities in the Use of Technology Tools and Resources Module Outcomes

1. Described the community of learners as citizens who share and utilize digital materials
2. Practiced standard netiquette in sharing and utilizing shared materials among learning communities
3. Identified examples of Intellectual Property Right in educational setting

Introduction

Technology has become an integral part of our lives. It has created a great impact that whether we like it or not, we have become prone to safety issues. Our privacy can be intruded; our social relationships can be affected. But it is not just enough that we keep ourselves safe, we should also possess the skills on how we can act responsibly so as not to pose harm to others. Thus, this Module will walk you through the issue.

Lesson 1 Digital Citizenship vs. Global Citizenship

Lesson Outcomes

1. Recognized the five tenets of global digital citizenship and the 9 elements of digital citizenship
2. Compared and contrasted one's role as a citizen of a community and that of a digital word 3. Shared ways on how one can observe social, ethical and legal responsibilities in the use of technology tools and resources.

Introduction

Are you a Filipino citizen? As a member of a smaller Filipino community such as a barangay or a subdivision, what are your rights and responsibilities?

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Do you use the internet? Are you a member of a cyberspace community? What are your rights and responsibilities as user of the internet?

Fill out the Venn diagram below for your answers.

My Rights and Responsibilities

As Filipino Citizen As Internet user

Discussion

Digital Citizenship vs Global Citizenship

Since we are considered citizens of the community we live in, then as users of the internet we can also call ourselves as citizens of the digital world where we live, learn and work in an interconnected manner. Digital citizenship is an idea that all persons using the internet have civic rights and responsibilities. It center around safe, savvy and ethical use of the technology.

A global citizen sees the world as a community in which all people live and prosper together. He/she understands that his/her actions contribute to the values of the entire planet and he/she is concerned on how he/she participates in and contributes to the entire world. A **digital citizen**, on the other hand, adheres to guideline that governs the ethical and responsible use of technology and acts responsibly in all relationships and interactions in the digital world.

When we mix these two together, it's the perfect recipe for the **Global Digital Citizen**. A global digital citizen is a responsible, ethical citizen leveraging technology to foster community on a global scale through connection and compassion. The Global Digital Citizen understands that we can govern technology for the benefit of both ourselves and others. It is a citizen that views the world as an interconnected community. Additionally, the digital citizen realizes that we simultaneously share technological and human experiences regardless of culture, status or political/religious beliefs (Watanabe-Crocket,2017).

Five Tenets of Global Digital Citizenship

We define the best assets of the global Digital Citizen using 5 tenets: Personal, Responsibility, Global Citizenship, Altruistic Service, Environmental Stewardship, and Digital Citizenship (Ribble,2017). These five tenets are the essence of what it means to be a great Global Digital Citizen.

Personal Responsibility

Personal responsibility includes demonstrating how we manage ourselves in matters such as personal finances, ethical and moral boundaries, personal health and wellness and relationships of every kind, both online or offline. For the student, this kind of responsibility means taking ownership of his/her learning. Developing a sense of accountability for lifelong learning in our students prepares them for the workplace. These are places where they must often think and act independently and take initiatives in their own. Encouraging this in the classroom as early as possible can help students develop high-level critical thinking and problem solving abilities to assist them in every aspect of their lives.

Global Citizenship

The Global Digital Citizen understands that technology has dissolved boundaries between all worlds' people. We now communicate, collaborate, and celebrate across all levels of society. We are now all global citizens. Our students must realize, among other things, that we aren't isolated from each other anymore. A deep understanding of this new global interconnectedness can encourage them to be aware of the issues, traditions, values and cultures of other citizens. This leads to cultivating understanding, acceptance, compassion and humility.

Digital Citizenship

Digital citizenship covers appropriate and exemplary behavior in ur online environments. It's about working towards making our transparent digital world safe for ourselves and others. Just like personal responsibility, this involves moving accountability for appropriate behavior we teach them how to govern their behavior, act in just and moral ways, and lead by good example in all online

associations. In doing so, they foster independence within themselves and hopefully light the way for others to do the same.

Altruistic Service

Altruistic is defined as "having a selfless concern for the well-being of others." The Global Digital Citizens acknowledge that they share this world with many different people. These ideals apply to the people we know, but also to those we don't know. These include embracing the opportunity to exercise clarity and goodwill for the benefit of others. The practice of altruistic service provides opportunities for our students to create meaningful connections to the real world. It gives them the chance to reach out to others and do things can make a person's experiences better or easier in everyday life. This produces a sense of community and well-being for all.

Environmental Stewardship

This practice is all about common-sense values and an appreciation for the beauty and majesty that surround us every day. This facet of Global Digital Citizenship encourages exploring how we can practice the conscientious use of Earth's resources. Environmental stewardship encourages every student to take a positive stand on personal, local, regional, national and international actions regarding the preservation of what is essentially our "environmental community". This facet of Global Digital Citizenship reinforces the idea that we have one world that we all live on. It's vital that we respect it and take steps to ensure its beauty and longevity for all generations (Watanabe-Crockett, 2017).

Why Do We Need to Be Global Digital Citizens?

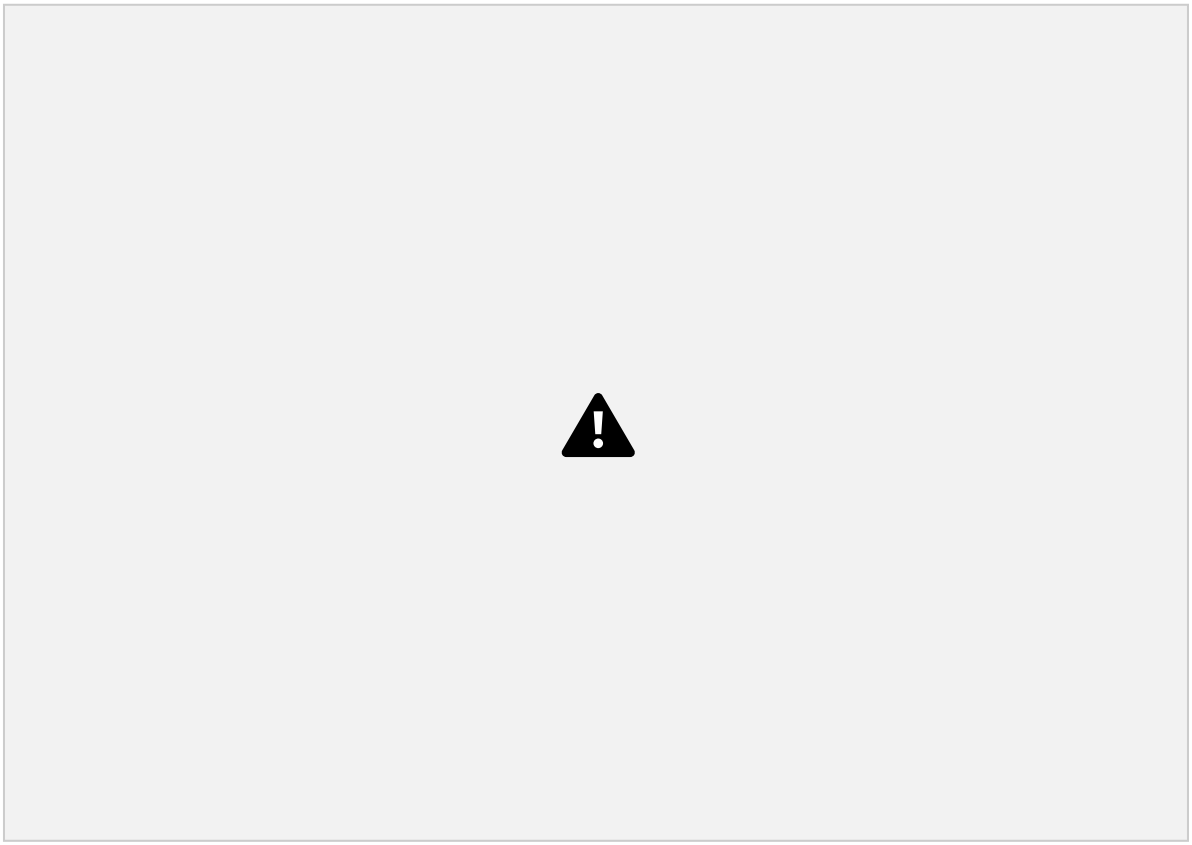
Here's an idea from Watanabe-Crockett(2017). **Personally,** we face the daily possibility of online fraud, identity theft and online bullying. We place our entire lives and their contents in the digital stratosphere willingly and often without considering the consequences. The dangers of this are real.

What is needed is people who will advocate and demonstrate the kind of self-governance that will keep us date from harm both online and offline. **Globally**, even as technologically intertwined as we are, there is also a level of disconnection in our lives. We are more connected than ever and yet often less tuned into others than ever. Global Digital Citizens see the need that all diversity has to be recognized, honed and cherished for present and future generations. They also see the need for harmony and connection in a world transforming rapidly as ours. **Digitally**, our internet is unchecked and growing wild, which is a good thing. Online information is expanding at a staggering rate. Online perils do exist, and the Global Digital Citizen sets an example to others on how to navigate them safely and securely. A good helping of common sense goes a long way in this regard. **Altruistically**, we have more means to help others. Thanks to technology. Crowd funding and crowdsourcing help us back to local and global communities. Social media can become a support network for those of us who are in physical, mental and emotional need. The Global Digital Citizens can lead the way.

Environmentally, we continue to face threats such as climate change, resource depletion, industrial pollution, digital pollution and more. Global Digital Citizens seek to combat the biggest challenges to our environmental health. They foster solutions that work to the benefit of the entire planet.

Elements of Digital Citizenship

Digital Citizenship is now a component of the digital and is in fact one off the facets of Global Digital Citizenship. Under digital citizenship, and is in fact one of the facets of Global Digital Citizenship, there are nine elements: digital access, digital commerce, digital communication, digital literacy, digital etiquette, digital law, a digital rights and responsibilities, digital health and wellness and digital securities.



https://www.thinglink.com/scene/753442770672156672

Digital Access

Equal digital rights and electronic access is the starting point of digital citizenship. Digital access is supposed to be available to all in the same manner that education is made sure by the Philippine government to be available to all citizens. Currently, on this matter, the Republic Act 10929 was signed on July 25,2016 requiring all public places in the country to have free internet access. Public places in this act covers national and local government offices, public basic education institutions, state universities and colleges and TESDA technology institutions, public parks and barangay reading centers public airports and seaports, and public terminals. Despite this law, however, many are still deprived of internet access and therefore deprived of some resources. To become productive citizens, we need to provide and expand access to technology and we need to be committed that no one is denied digital access.

Digital Commerce

Technology users need to understand that a large share of market is being done electronically. Nowadays people shop through different websites and even do online transactions for banking. Many other online transactions are occurring including the illegal downloading, pornography and others thus a safe and secure online shopping and other commercial transactions have to be done. With this, an e commerce act or RA 8972 was signed in July 2000. This Act aims to facilitate domestic and international dealings, transactions, arrangements, agreements, contracts and exchanges and storage of information through the utilization of electronic, optical and similar medium, mode, instrumentality and technology to recognize the authenticity and reliability of electronic documents related to such activities and to promote the universal use of electronic transaction in the government and general public.

Digital Communication

It is another element of Digital Citizenship that we seem to be embracing wholeheartedly. Digital communication is the electronic exchange of communication which includes emailing, texting, instant messaging, utilizing cell phones, etc. As digital communication continue to grow, it is important that we do not forget how to communicate face to face. We have to utilize digital channels of communication to enhance our ability to communicate, but do not allow it to replace our ability to communicate.

Digital Literacy

As digital citizens, it is our responsibility to develop and continually enhance our technological knowledge. We should support and encourage the acquisition of technological knowledge by others. We should model the positive and proactive use of technology for good causes, personal and professional growth and education.

Digital Etiquette

This refers to the electronic standards of conduct or procedure. This is often seen as one of the most pressing problems when dealing with Digital Citizenship. Imagine a teenager who spreads hate speech about a classmate online. Or none who browses Facebook while attending Mass. Browsing social media sites have become so addictive that young people tend to lose distinction between what is appropriate and what is not. It begins with rules . It is not enough however to create rules and policies, everyone must be taught to become responsible digital citizen.

Digital Law

Digital law refers to the electronic responsibility for actions and deeds. The digital word is so huge that there should be order, discipline and ethical use. Plagiarism, copyright infringement, hacking of information, identify theft are crimes that are commonly violated by online users. As digital citizens, we must be aware of these illegal behaviors so that we can avoid them and report those who illegally engage in them.

Digital Rights and Responsibilities

"For every right, there is a corresponding duty and for every privilege, there is a corresponding responsibility." The privileges of digital access come along with responsibility. Just as the Philippine constitution protects us with rights of being Filipino citizens such as the right to privacy, we also have responsibility to be cautious in our online activities such as posting our photos and video online.

Digital Health and Wellness

Our physical and psychological well-being in a digital technology world should be a priority. Many illnesses have become associated with technology use such as carpal tunnel syndrome, eyestrain and childhood obesity as an effect of children's prolonged sitting in front of the computer screen. Technology has also posed an inherent danger of addiction and threat from online predators. Thus, children and adults alike must learn how to protect themselves as digital citizens through education and training.

Digital Security

How to keep oneself secure in any community is a major concern. If we keep our homes safe with gates and locks, we also need to protect the information that we provide online with privacy setting and strong passwords. We also need to have virus protection and backups of our data. As responsible digital citizens, we must know how to keep pour data, our money and our identity from all hazards in the digital world.

Digital citizenship is very similar to our citizenship in our immediate community. There are social, ethical, and legal issues involved in our day to day dealings with people and the surroundings. It is not just about recognizing and dealing with online hazards. It is about building safe places and communities, understanding how to manage personal information, and about being savvy - using our online presence to grow and shape our world in a safe, creative way, inspiring others to do the same (Blackboard, 2018). As a global digital citizen, we need to be conscientious and compassionate; we should strive to establish a sense of global community in all our online and offline relationships, duties and endeavors (Global Digital Citizen Foundation).

To Do!

Reflective Essay 6.1: In writing your reflection always use Gibb's Reflective Model and look back at how you engage in activities online, what kind of digital citizen were you?

Lesson 2

Netizenship and Netiquette in Online Communities

Lesson Outcomes

1. Defined netizenship and netiquette
2. Formulated a set of netiquette guidelines

Introduction

Study the case below and analyze the side of both parties.

The case involved two minor students from a certain school whose photos were posted on Facebook. The photos, which were uploaded by one of their friends showed the students drinking and smoking in a bar and wearing just undergarments on a street. The photos were shown them to ban the students from marching in their graduation rites. According to the school, the students violated the school code of conduct.

The parents of the students in defense filed a petition for the issuance of a writ of habeas data and asked the court to order the school to surrender and deposit all soft and printed copies of the photographs and to declare they have been illegally obtained in violation of the children's right to privacy.

After weighing in all the information, with whom will you side-to the parents or to the school?

Read the Supreme Court's decision.

The court dismissed the parent's petition and ruled that, "The school did not violate the minors' privacy rights." According to the court, the school cannot be faulted for being "steadfast in its duty of teaching its students to be responsible in their dealings and activities in cyberspace, particularly in (social; networks), when it enforced the disciplinary actions specified in the Student Handbook, absent a showing that, in the process, it violated the students' right." The decision of the court stated that the students cannot invoke the protection attached to the right to informational privacy because the photos were seen by other student of the school, who in turn showed them to the computer teacher who reported the incident to the school authorities. In the language of the court. "The photos, having been uploaded on Facebook without restrictions as to who may view them, lost their privacy in some way." The court further added in its ruling that setting their post privacy to 'Friends Only' is not an assurance that it can no longer be viewed by another user who is not a Facebook friends with the source of the content. The decision read that, "Without proof that they placed the photographs subject of this case within the ambit of their protected zone of privacy, they cannot now insist that they have an expectation of privacy with respect to the photographs in question.

The decision of the court puts the burden of ensuring safeguarding privacy online users and expects them to exercise due diligence the role of schools and parents in disciplining and educating their children to be a good digital citizens, self-regulation is the "best means f avoiding privacy rights violations." It further issued a ruling saying that nothing is ever private on Facebook; even those tagged as private never really escape public viewing, including unintended audiences.

The Philippines, being considered as asocial media capital of the world, has a need to stress social media education and etiquette among online users. The interview conducted with parents and

teachers of high school students in the above example underscored the need to use social media responsibly. All agreed for the inclusion of social media etiquette in the education curriculum either as separate subjects or integrated in good manners and right conduct. Social media can attain many things in terms of social good (Brutas, 2015). How can social media be used to affect positive change?

While working online we all create a digital tattoo and we lever digital footprints. We have to remember that our digital world is permanent and with each post, we leave a digital footprint. By doing self-reflecting before we self-reveal, we are able to consider how what we share online can impact ourselves and others (www.commonsensemedia.org).

Scrutinizing the term again, a netizen, according to the Merriam-Webster Dictionary (26 April 2018) is n active participant in the online community of the internet. The medical dictionary refers to it as an internet citizen who uses networked resources, which connotes civic responsibility and participation. Thus netizenship means citizenship ion the internet or in the virtual world.

As responsible netizens, we are all expected t support a healthy interaction on the internet. A netiquette or the rules of socially accepted behavior online have to be observed in writing an email, in texting and other communications on the web.

Below are some netiquette guidelines:

1. **Protect your reputation.** Whether in real or virtual world, you are the same person. Do what is only appropriate and share information that does not harm you as a person.

2. Respect others. Respect begets respect. Treat everyone with respect even if you have not seen him/her in person. Be judicious about what you say on your own and other's pages.

3. Express yourself clearly and use emoticons. Communication online is difficult because emotions are not evident during communication. Miscommunication usually takes place because yurt facial expression and your body language cannot be seen and the tone of your voice cannot be heard. Thus, emoticons are readily available to show your emotions.

4. Remember the intellectual property. Ideas online are products of intelligence of others. If you need to cite them, acknowledge the authors. You definitely do not want to steal

properties of others.

5. Check spelling, grammar and punctuation. Since your face cannot be seen online, you will be judge according to your posts. Good writing means good manners. We do not want to waste other people's time reading our post which is incomprehensible.

6. Pause before you post. Take note that whatever you post become permanent therefore think twice or thrice before you click send.

7. Do not share your personal information. Sharing your personal information online is like going around streets wearing a shirt printed with your name, birthday, address, name of

parents, etc. You definitely do not want to reveal all these to many people. Besides, providing all these publicly will make you prone to identity theft.

8. Think about who or what you are representing. As a son/daughter, you represent your family. As a student, you represent your school. You do not want your family or school to be put to shame by what you do.

Each of us has a significant contribution to the kind of community we have in the virtual world. As the Association of American Medical Colleges (2015 in www.cyberbullying.org) say it, "We are responsible for the 'public face' that the connected world sees". Consequently, our individual digital reputation is shaped by our posts on social media.

To Do!

Reflective Essay 6.2: In writing your reflection always use Gibb's Reflective Model and look back on a situations or dilemmas you encountered regarding online behavior. Provided a netiquette, how will you practice it?

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Technology for Teaching and Learning 1

LESSON 3

Intellectual Property Rights on the Development and Use of Digital

Materials Lesson Outcomes

1. Identified examples of Intellectual Property Right in educational setting
2. Defined the copyright laws

Introduction

Plagiarism.org reported that…

A survey of over 63,700 US graduates and 9,250 graduate students over the course of three years (2002-2005) – conducted by Donald McCabe, Rutgers University – revealed the following:

1. 36% of undergraduates admit to "paraphrasing/copying few sentences from Internet source without footnoting it"

2. 24% of graduate students self-report doing the same
3. 38% admit to "paraphrasing/copying few sentences from written source without footnoting it"
4. 25% of graduate students self-report doing the same
5. 14% of students admit to "fabricating/falsifying a bibliography"
6. 7%of graduate students self-report doing the same
7. 7% self report copying materials "almost word for word from a written source without citation"
8. 4% of graduate students self report doing the same
9. 7% self-report "turning in work done by another"
10. 3% of graduate students self-report doing the same
11. 3% of report "obtaining paper from term paper mill"
12. 2% of graduate students report doing so

The Josephson Institute Center for Youth Ethics surveyed 43,000 high school students in public and private schools and found that: One out of three high school students admitted that they used the Internet to plagiarize an assignment

Discussion

Intellectual Property Right

Intellectual property protection is important in fostering innovation. Without protection of ideas, individuals will not be able to enjoy the full benefit of their inventions and they would not be compensated for their creations. This encourages innovation without the fear that a competitor will steal the ide and/or take the credit for it.

Intellectual property, according to World Intellectual Property Organization (WIPO), refers to creations of the mind such as inventions; literary and artistic works; designs; and symbols, names and images used in commerce. An intellectual property right is right held by a person or by a company to have exclusive rights over these. Internationally, these intellectual properties are protected by the World Intellectual Property Organization to which the Philippines is one of the 191 member states. In our country, we have Intellectual Property of the Philippines (IPOPHIL) which administers and

implements state policies in relation to intellectual property. This office was created through Republic Act 8293. This act defines intellectual property to include copyright and related rights; trademarks and service marks; geographic indications; industrial designs; patents; layout designs (topographies) of integrated circuits; and protection of undisclosed information. In the education setting, copyright and related rights are the ones applicable.

Copyright

Copyright refers to the legal right given to the owner of the original work or intellectual property. These "works" are original intellectual creations in the literary and artistic domain protected from the moment of their creation which include the following:

1. books, pamphlets, articles and other writings;

2. periodicals and newspapers;

3. lectures, sermons, addresses, dissertations prepared for oral delivery, whether or not reduced in writing or other material forms;

4. letters;

5. dramatic or dramatic-musical compositions; choreographic works or entertainment in dumb

shows;

6. musical compositions, with or without words;

7. works of drawing, painting, architecture, sculpture, engraving, lithography or other works of art, models or designs for works of art;

8. original ornamental designs or models for articles of manufacture, whether or not registrable as an industrial design, and other works of applied art;

9. illustrations, maps, plans, sketches, charts and three-dimensional works relative to geography, topography, architecture or science;

10. drawings or plastic work of a scientific or technical character;

11. photographic works including works produced by a process analogous to photography; lantern slides;

12. audiovisual works and cinematographic works produced by a process analogous to cinematography or any process for making audio-visual recordings;

13. pictorial illustrations and advertisements;

14. computer programs; and

15. other itinerary, scholarly, scientific and artistic works.

The internet is a very rich source of these materials and we should be careful in using them. These works are protected by the sole fact that their creation, irrespective of their form or mode of creation as well as of their content, quality and purpose. Now that copy and paste are very accessible commands in our computer toolbars, we should try not to violate the copyright laws otherwise we could be accused of copyright infringement. Copyright infringement is the use of works without permission where the copyright holder has the exclusive right to reproduce, distribute, display or perform the protected work, or to make derivative works. Proper citations should be used when including them in our output.

Below are the guidelines on online use of copyrighted materials by Smaldino, Lowhter and Russel (2012):

1. Contrary to the popular opinion, all materials in the internet is copyrighted unless stated otherwise. It is copyrighted even if it does not display the copyright symbol.
2. An email is an original work., fixed in a tangible medium of expression, that is covered by copyright. Hence it is recommended that you should not forward any email without permission, in consideration of both copyright and Privacy Act.
3. Downloading an article from a newspaper's website, making copies, and distributing them to your students prior to a class discussion on the topic is permissible following the current

photocopying guidelines which permits making multiple copies for classroom use. The exception would be individually bylined, copyrighted articles or articles from a source specifically designed for the educational market where such articles cannot be copied legally for class distribution (adapted from Becker,2003)

4. You cannot post students' essays, poems, or other works on the school website unless you have permission of the students and their parents or guardians.
5. Educators should treat copyrighted materials from the internet the same way they do to print formats. The best guideline is to always obtain permission. When in doubt, ask!

Copyright Infringement vs. Plagiarism

Another violation on intellectual property is plagiarism. Although plagiarism and copyright infringement are related ideas, these two are different. Plagiarism, according to plagiarism.org, is an act of fraud; it involves both stealing someone else's work and lying about it afterward. Plagiarism.org further elaborates the following as plagiarism:

1. turning in someone's work as your own
2. copying words or ideas from someone else without giving credit
3. failing to put quotation in quotation marks
4. giving incorrect information about the source of quotation
5. changing words but copying the sentence structure of a source without giving credit 6. copying so many words or ideas from a source that it makes up the majority of your work, whether you give credit or not (see our section on "fair use" rules).

For images, videos and music, the following are counter as plagiarism:

1. copying media (especially images) from other websites to paste them into your own papers or websites.
2. making a video using footage from other's video or using copyrighted music as part of the soundtrack.
3. Performing another person's copyrighted music (i.e. playing a cover)
4. Composing a piece of music that borrows heavily from another composition.

In other words, plagiarism refers to copying the work of another and claiming it as one's ideas or without p[roper attribution while copyright infringement is copying one's work without obtaining permission. In this sense, copyright infringement is a violation of the right of the copyright holder while plagiarism is a violation of the right of the author. We have to note that not all authors are copyright holders. There are cases when the publisher owns the copyright. In addition, copyright infringement is a legal violation while plagiarism is an ethical violation.

To Do!

Reflective Essay 6.3: In writing your reflection always use Gibb's Reflective Model and as a pre service teacher, cite situations that will ensure that you are not violating any laws of copyright infringements and plagiarism.

"You cannot fail when you know you've done your best. You may either succeed or learn in the process. Either way, you should feel good about yourself. What matters is you've given your all."

-Myke Celis

End of the module.....

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